

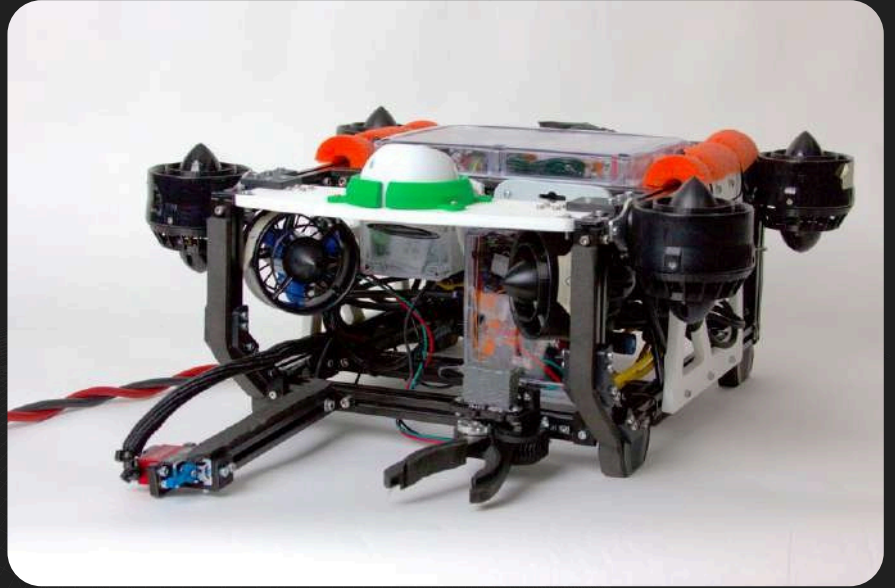


MUREX ROBOTICS

YASH SHAH, PHILLIPS EXETER ACADEMY
>> EXETER, NEW HAMPSHIRE, USA

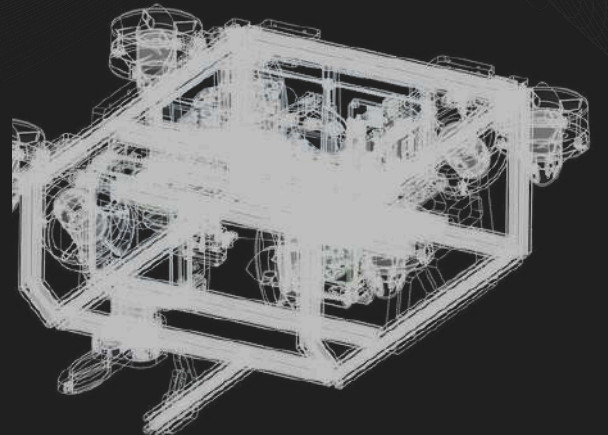
// ATTEMPT THE IMPOSSIBLE.

Yash's Maker Portfolio



Disclaimer: Parts of portfolio are adopted from MUREX's technical report for 2023-2024 season.

YashShah [ChiefProductDesignOfficer,DriveTeam](1st Year/'25)



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ABSTRACT

This year, I, led MUREX (MATE Underwater Robotics at Phillips Exeter) Robotics Company of Phillips Exeter Academy (PEA) from Exeter, New Hampshire, USA to engineer and assemble the MUREX ROV V3.5, a small, lightweight, low cost, modular, and fully open source ROV to meet the challenges described in the 2024 Marine Advanced Technology Education's (MATE) Request for Proposals (RFPs) and to address the needs of the global marine robotics community. The MUREX ROV V3 and crew work to: 1. Collect and analyze oceanic data, 2. Manipulate and maneuver undersea cabling 3. Conserve ecosystems and save wildlife, 4. Engineer, construct, and demonstrate a vertical profiling float (mrxFLOAT).

*The MUREX ROV V3.5 is designed for challenging environments following my **3 years of iterative robotics design experience**. This robot is fully custom-designed to meet our customers' standards while being completely open-sourced to further ROV development in the underwater robotics community.*



>> CHALLENGES

In following my mission to attempt the impossible, it is easy for me to get carried away with too many ambitious goals. I've researched and attempted many ideas, but most remain on the drawing board. Deciding whether or not to put time and effort into an idea is incredibly difficult. It is a constant challenge I tackle every day, from small decisions that might save us a few minutes, to large overhauls of the robot. Decisions are split into three parts: ideation, research, and conclusion. When an idea is created, I conduct "viability checks" asking:

- "What issue is it trying to solve?"
- "How long does it take?"
- "How much work would it take?"
- "How much money does it cost?"
- "Does the benefit outweigh the downsides?"



B-Roll of the MUREX Ethernet Switch V1.0

After careful consideration, a conclusion is reached. This process can be as fast as one online search to a multi-day journey of reading whitepapers or learning new knowledge.

>> ***MUREX Robotics is a fully student-led and student-developed team.***



MUREX Ethernet Switch V1.0



December Progress Report.



// INNOVATIONS

Problem: Lack of low-cost, advanced electronics for underwater robotics

Solution: One of our biggest breakthroughs is the MUREX electrical system. I focus on making fully open-source electronics that undercut commercial offerings at 5-10% of the cost while being interoperable with existing options. (See Control/Electrical System)

My team and I are proud to have achieved the following:

- [Dec. 2023] World's first open-source Ethernet Switch, MUREX Ethernet Switch
- [Mar. 2024] World's first open-source CM4-based ROV control board, MUREX Carrier Board
- [Mar. 2024] World's first open-source implementation of a USB3.0, MUREX Carrier Board controller
- [Nov. 2023] World's first modular serial single ESC, MUREX ANYESC
- [Dec. 2023] World's smallest open-source dual-output high power module designed for 12V/5V/3.3V DC applications, MUREX Power Board
- [Nov. 2023] World's first open demo of OFDM (orthogonal frequency division multiplexing) over galvanically isolated DC power lines, mrxPLC
- [Jan. 2024] World's first use of H.264 hardware encoding on the Allwinner H3 SoC using a mainline Linux kernel, unnamed
- [Dec. 2023] Novel, direct-compilation of the Linux Kernel with a virtualized Apple Silicon macOS environment and minimal overhead, unnamed
- [Apr. 2024] Xbox trigger support for Ardupilot + QGroundControl, unnamed

ALL DEVELOPMENTS ARE OPEN SOURCED ON GITHUB

// SYSTEMS APPROACH

My team and I have designed the robot's mechanical, software, and electrical systems in parallel.

- Software infrastructure for the electrical team to design around
- Standardized electrical connectors between the PCBs
- Enclosure dimensions for the mechanical team to design around

This allowed our team to work asynchronously while assuring each component of the robot could be assembled and function as expected. This was a major improvement to our scheduling and planning in comparison to last year, where the software design relied on the completion of the electrical system, which relied on the mechanical design.

Each system was designed with **adaptability in mind**. The mechanical design offers maximum flexibility while providing a streamlined shape to minimize drag. T

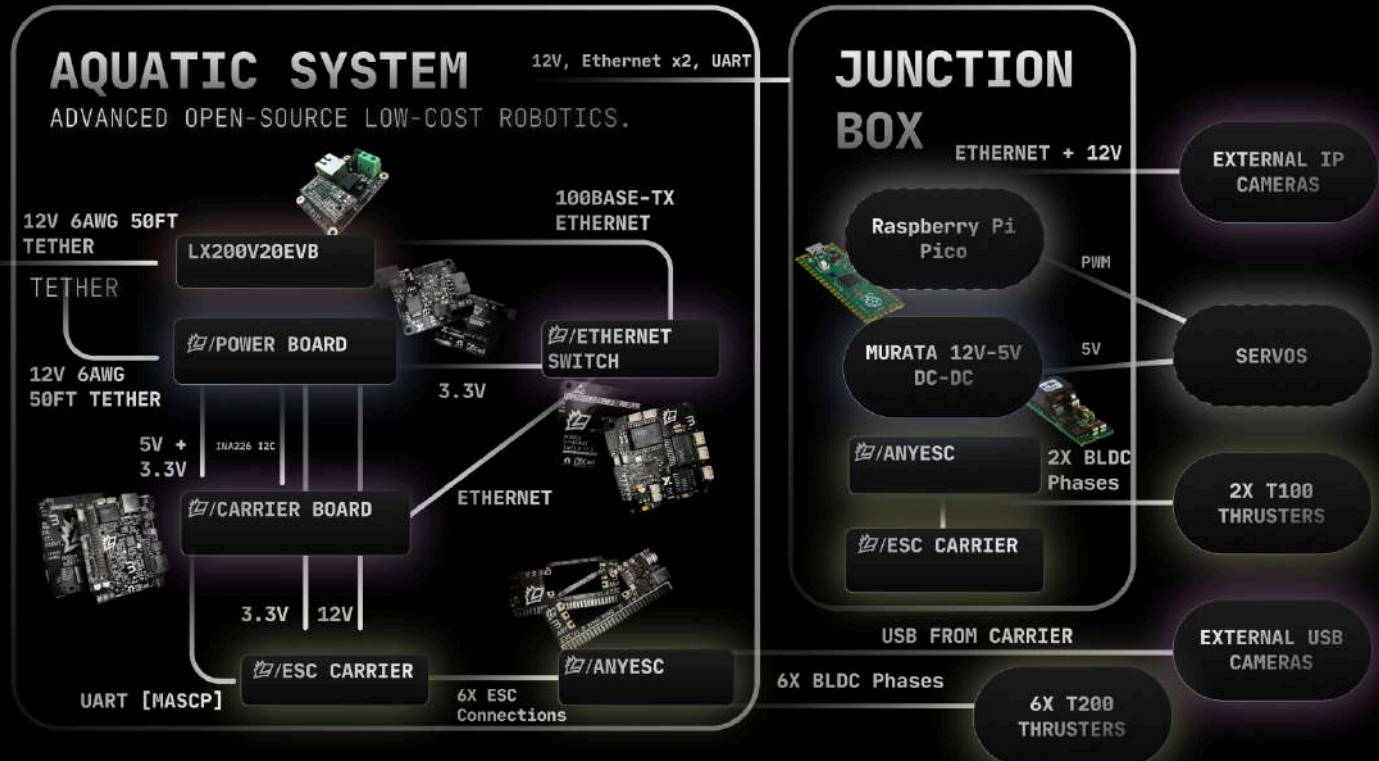


Experimenting with different epoxy brands and compositions while waterproofing our servo.

CONTROL/ELECTRICAL SYSTEM

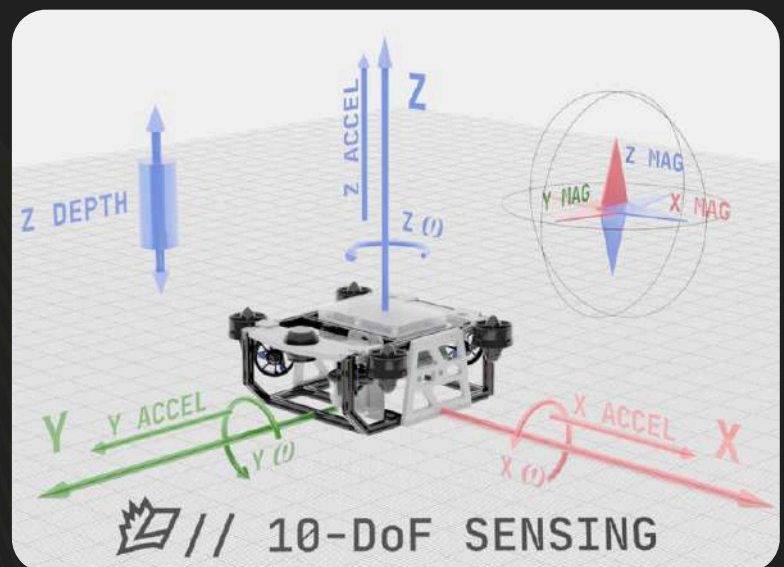
MUREX INTEGRATION DIAGRAM

ATTEMPT THE IMPOSSIBLE. DO IT RIGHT. THINK DIFFERENT.



> OVERVIEW

I've designed the electrical system to be **open-source that offers 10-DoF sensing, input power, transient protection** on every major connection. I've utilized a custom Linux distribution, mrxOS, a custom control software built on top of Ardupilot, mrx, and many custom software systems such as MASCP (MUREX Asynchronous Serial Communication Protocol) and custom scripts for Ardupilot. A optimized software in combination with the advanced electrical system enable the MUREX ROV V3.5 to glide through water and reliably carry out tasks such as placing brain coral, enabling irrigation systems, and deploy subsea cabling.



> CARRIER BOARD

The MUREX Carrier Board V3 is an ROV control system featuring the world's first **open-source CM4-based ROV control board and the world's first open-source implementation of a USB3.0 controller in any form**. USB3.0's 5Gbps speed is currently used for video input but can expand to networking, AI acceleration, storage, and high-speed DAQ systems. The board costs ~7% of alternative commercial options, with our prototyping run costing around 30 dollars per board, and offers a suite of features previously inaccessible at this price point. Although there are more powerful System on a Chip (SoCs) options from Rockchip and Allwinner, I chose to continue development in the Raspberry Pi ecosystem for the software support and electrical reference material [3]. **Features include:**

- A controller supporting the USB 3.0 and xHCI specifications with the uPD720202 [4].
- A dual-chip design with an onboard Espressif ESP32-S3 wireless and GPIO capabilities [5].
- BMI088, LIS3MDL, BME680, MS5837 sensors for 10-DoF sensing (XYZ acceleration, XYZ angular velocity, XYZ spatial orientation, pressure/water column depth).
- 2 TPS259474LRPWR eFuses offer independent input protection and monitoring of the 5V and 3.3V power inputs.

> POWER BOARD

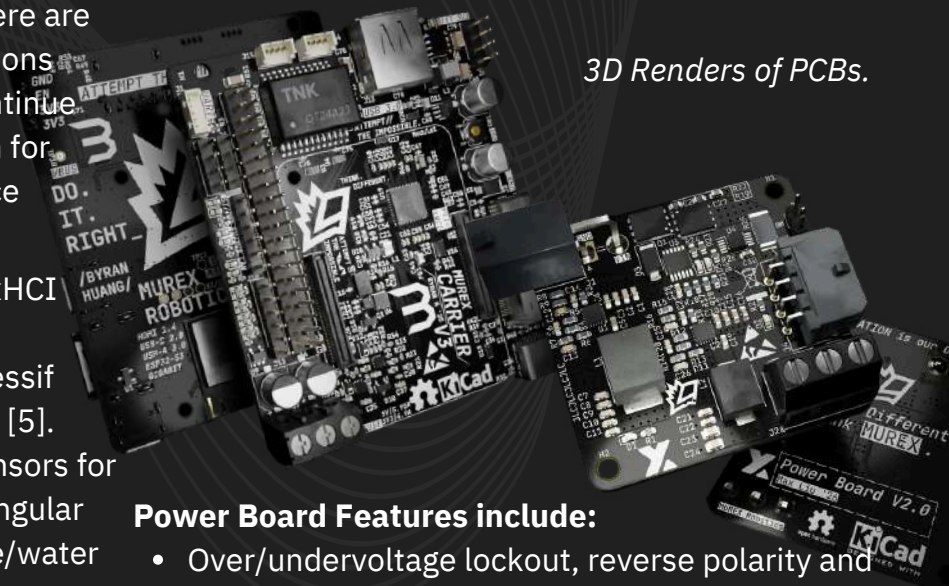
The MUREX Power Board V2 is the **world's smallest open-source dual-output, high-power module designed for 12V/5V/3.3V DC systems and ROV applications**. Measuring 44mm x 46mm, the board supports 500W continuous power and is 95% efficient at DC-DC conversion. This is used as our power input and DC-DC converting system, providing safe voltage levels to every part of the system.

Notes: [2] Please see build/buy section. [3] I am experimenting with embedded Linux to become the one of the first people to accomplish hardware H.264 encoding on Allwinner systems as opposed to simpler but less performant software encoding. [4] See the world's only active and purchasable [USB3.0 controller IC](#). [6] Referenced **TI white papers** SLVAED3, SLVAEG8, SLVA882B and more.

> OVERVIEW CONT.

The system communicates through MASCP to the MUREX ANYESCs, which house commercial ESCs and drive T200 and T100 thrusters. The ANYESCs slot into the ESC Carrier for modular storage. Finally, a Murata 12V-5V buck converter provides up to 20A for powering external servos for the arm/manipulators, which receives PWM from a dedicated Raspberry Pi Pico. Video is streamed from IP and USB cameras through the Ethernet switch, the mrxPLC system, then decoded into Ethernet in the topside.

In total, the MUREX ROV V3.5 has over 13 custom PCBs, enabling various functionality at costs <12% of existing commercial options [2]. MUREX will now provide further descriptions and functionalities for custom boards and methodologies in use.



3D Renders of PCBs.

Power Board Features include:

- Over/undervoltage lockout, reverse polarity and shorting protection, in-rush current limiting, and transient spike protection with the LTC4352 Ideal Diode Controller.
- Current and voltage monitoring with the INA226 Power Monitoring IC.
- Beat-frequency correction, high-power, and efficient dual DC-DC conversion with the TPS568230 Synchronous Buck Converter [6].

> ETHERNET SWITCH

The MUREX Ethernet Switch V1 is the world's **first open-source ethernet switch**. With a focus on minimizing physical footprint and cost, the MUREX Ethernet Switch utilizes external magnetics, a high-performance unmanaged switch IC, and 1.25mm pitch Fast Ethernet (100BASE-TX) connectors. It costs 5% of alternative options and acts as the central communications center of the MUREX ROV, combining the 2 IP cameras, the LX200V20, and the MUREX Carrier Board. This board was developed out of necessity as full-size Ethernet switches were too large to fit into the system and commercial options were priced extremely high.

Features include:

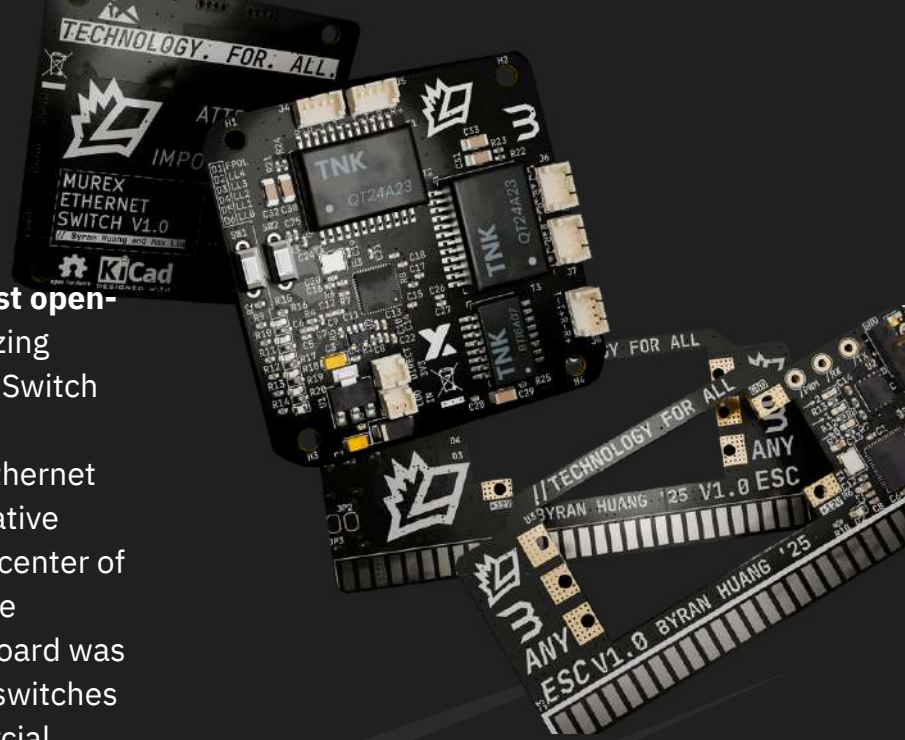
- 5 802.3u compliant Ethernet ports with the IC Plus IP175G [7].
- 4.5 - 15V protected input through the LM1117MP-3.3 800mA LDO.
- Direct 3.3V power input for high efficiency.

> mrxPLC AND TETHER ELECTRICAL

MUREX has conducted extensive research and development on a novel galvanically isolated orthogonal frequency division multiplexing (OFDM) system across the power tether, **allowing our robot to have over 30-60 MBits of data bandwidth with only power and ground on our tether**. A system explanation diagram is provided in the Topside System section. I have named this development the MUREX Power-Line Communication System (mrXPLC). It enables the robot to use only two wires for the tether, minimizing cost and possible damage. This system is even more effective for high-voltage tethers.

> CONNECTORS AND WIRING

To comply with ELEC-017R, all wiring is standardized into either: Molex Picoblade/Header Pins, Molex Mini-Fit, WAGO, or heatshrinked. All data lines are Picoblades or 2.54mm header pins for their compact size. All high power connections are Molex Mini-Fit or WAGO connects if on a high power bus (like main 12V). Notes: [7] See [this](#). [8] See [this](#). [9] See [this](#).



> ANYESC + ESC CARRIER

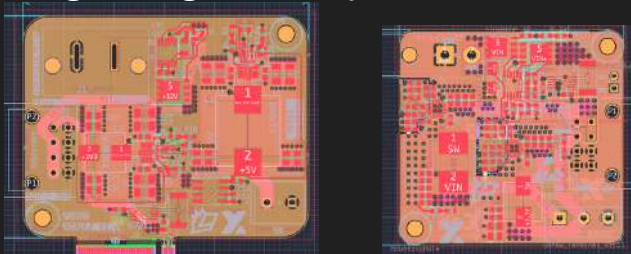
The MUREX ANYESC V1 is the **world's first modular serial single ESC** and our first prototype ESC for the MUREX Motion System. The ANYESC integrates commercial ESCs into a flexible RP2040 board in a card-edge form factor (hence the name “ANY”). The RP2040 was chosen for its balance between optimal price, performance, power consumption, and more. It acts as an extremely powerful translator, supporting UART and PWM communication to any ESC mounted [8]. Adaptable length for any commercial single-sized ESC is also supported. MUREX has also conducted testing with field-oriented control of our thrusters with our concept MUREX DRVESC V1 design. The MUREX ESC Carrier V2 integrates the world’s first open-source serially connected motion system. In-rush current limiting with an NTC thermistor Standard high-power Molex connectors for fast replacements

ANYESCs can slot into the MUREX ESC Carrier, with the RP2040 ensuring full compatibility no matter what ESC is slotted (allowing for future custom high-power or FOC ESCs to be implemented with no hardware modifications).



Left to Right: Picoblade, Mini-Fit (Molex), WAGO 221 (WAGO)

Carrier V1 -> V3 | 42% area decrease.



All files - MUREX

Carrier Board V1.1

- [Recents](#)
- [Starred](#)

Name
HDMI Successful.JPG
Carrier V1.1 #1.HEIC
Carrier V1.1 #1 2.HEIC
4ch I2C Test.MOV
3rd I2C Test.MOV
2nd Naopixel Test.MOV
2nd I2C Test.MOV
1st Test.MOV
1st test with CM4.MOV
1st Test with CM4 Continued.MOV
1st Neoskel Test.MOV
1st I2C Test.MOV
1st HDMI Test.mov
1st Ethernet Test.MOV

Fix[carrier]: added bmi088 i2c addresses
Hello2999991 committed 6 months ago

Fix[carrier]: switch bmi088 to i2c mode
Hello2999991 committed 6 months ago

Commits on Nov 28, 2023

Feat[carrier]: v3 tape-out
Hello2999991 committed 6 months ago

Feat[carrier]: 99% taped out, silkscreening done
Hello2999991 committed 6 months ago

Feat[breakouts]: Fixed some ToDOs
Crane_Lea committed 6 months ago

Commits on Nov 27, 2023

Feat[power]: footprints assigned, initial layout
vamaU committed 6 months ago

Feat[carrier]: component mapping complete
Hello2999991 committed 6 months ago

Feat[carrier]: routing complete
Hello2999991 committed 6 months ago

Feat[power]: v2 schematic draft complete
vamaU committed 6 months ago

Commits on Nov 26, 2023

[illegible]

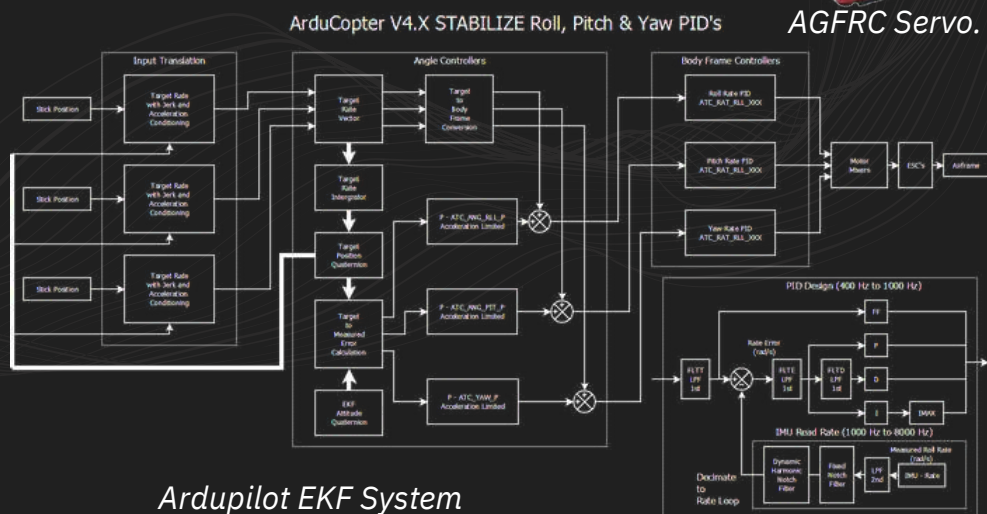
1. Research and determine main components.
2. Design your schematic (KiCAD eeschema)
3. Design your PCB (KiCAD pcbnew)
4. Design your silkscreen (KiCAD pcbnew)
5. Tapeout verification with the team (Discord)
6. Production with Fabricator (JLCPCB)
7. PCB Testing (mrxEE)
 - Incremental testing of every feature. Every test must be recorded and marked. (Dropbox)
8. Update documentation and generate renders. (Web Dev + Blender Visualization)

Through this process, I use GitHub to version manage as well as share our work with others in the underwater ROV community. Again, our work is fully open-sourced and documented.

VEHICLE SYSTEMS

CONTROL SYSTEM

The control system accurately controls the 8 thrusters and manipulator servos, allowing the demo team to make real-time observations about the mission. Our tuned EKF (extended Kalman Filter) with **22 estimation states across our 10 DoF** sensing allows our robot to be fully stabilized along all axes of motion.



MOTION SYSTEM

The motion system involves driving the 8 BLDC thrusters and controlling all servos. A major factor I considered this year was the complexity of existing ESC systems and their interoperability. I designed MUREX Motion System with **universalization in mind**, allowing for rapid expansion and modification of the system with minimal downtime and modifications.



AGFRC Servo.

T200 Thruster.



Enclosure Photo.

VISION SYSTEM

The vision system integrates our two IP (internet protocol) cameras and our main USB camera to stream live video to the topside. There is also a Logitech C270 upwards-facing camera. The MUREX Vision System includes a top-mounted, wide FOV camera and a lower, angled narrow FOV camera from Amcrest Technologies. These cameras were chosen for their **low cost, robust enclosures, and H.264 compression**. Many existing vision system implementations use USB cameras with multiplexing. However, that means only one camera can be displayed at a time. This is an inherent issue with video encoding on low-performance hardware, so MUREX bypasses the issue completely with our IP-based system. Our robot has three concurrent camera views, with the main USB camera (hardware encoded on the Carrier Board) providing the best clarity and lowest delay. The three cameras have intersecting FOVs, minimizing parallax. Commercial IP cameras also use robust outdoor enclosures, often made of cast aluminum with included O-rings, rubber seals, and desiccant. MUREX furthered the cameras' existing waterproofing with Molykote 111 and O-rings on each screw entry for further waterproofing. The ethernet outputs of the IP cameras connect to the MUREX Ethernet Switch.



Amcrest Camera.

Brio 101 USB Webcam.



MECHANICAL DESIGN RATIONALE

The MUREX ROV V3.5 (0.65m x 0.55m x 0.3m, 11.5kg) is made of 20 x 20 mm anodized aluminum extrusions, a standard size that has readily available parts and offers a **rapid building ecosystem** due to the infinite mounting capabilities of aluminum extrusions and T-slot nuts. The aluminum extrusions create a **large exoskeleton-like chassis** that is further reinforced with recycled marine-grade HDPE. The side HDPE sheets are pocketed to reduce water resistance in all directions. The HDPE plates also offer mounting for the Main Enclosure, Junction Box and four horizontal thrusters.

The vertical thrusters are mounted on both sides of the chassis on custom laser-cut aluminum parts which serve as rigid, corrosion-resistant mounting plates that are securely fastened to the extrusion.

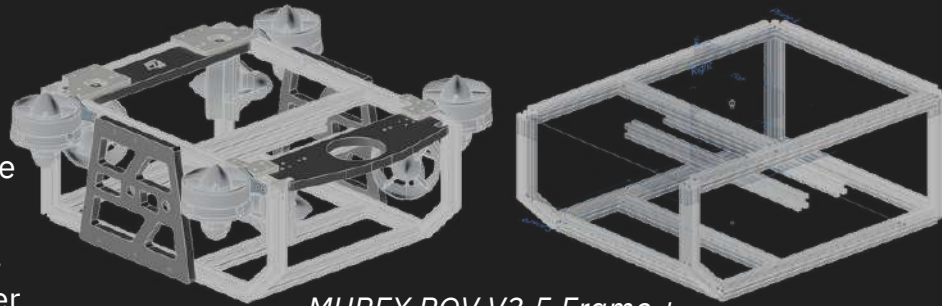
The front of the robot is angled to provide a wider reach of the manipulators and a better view from the vision system. It also acts as a visible marker of the direction of the robot.

The Main Enclosure is mounted in the center of the robot to provide balance in the water.

The area underneath the Main Enclosure allows for easy wire management and modifications. The Junction Box (Polycase ML-46F) is mounted vertically across the front to compensate for the tether's weight in the back of the robot.

MUREX ROV 3.5 is neutrally buoyant allowing for stable depth hold.

This provides reliable successes with mission tasks requiring a stationary robot such as activating the irrigation system that will protect and restore ecosystems and biodiversity.



MUREX ROV V3.5 Frame + Chassis CAD.

Chassis Comparison Table

Chassis	Material	Cost	Complexity	Benefits	Concerns
Aluminum extrusion exoskeleton	Anodized aluminum	Low	Low	High extensibility strength	ROV is limited by the extrusion's size and design
HDPE/Metal exoskeleton	Marine-grade HDPE/Aluminum	Medium	Medium	Very high strength Streamlined design	Cannot easily modify design
3D printed design	PLA	Low	Medium	Very high flexibility	Very low strength
Tube design	PVC/Carbon Fiber	Low or High	Medium	High strength and flexibility	Difficulty mounting components securely.

Enclosure Comparison Table

Enclosure	Size	Depth	Features	Concerns
Polycase Polycarbonate Enclosure	Large	Low (structural deficiencies beyond 50000 Pa)	Easy to design, mount, and repair. Transparent top for viewing.	Not optimal for underwater use, especially at depths.
Blue Robotics Cylindrical Enclosure	Small	High	Can withstand depths, designed for ROV usage	High cost, difficult to repair and troubleshoot.
Metal Enclosure	Large	High	Easy to design, mount, and repair.	High cost, not visibly inspectable.
Custom Enclosure	Flexible	Unknown	Flexible to match requirements	High risk of catastrophic failure.

I chose a *rectangular enclosure (Polycase ML-70F)* rather than a commonly used cylindrical enclosure, for reasons presented in the table to the left:



MUREX ROV V3.5 Main Enclosure CAD

BUOYANCY MODULE (mrxFLOAT)

The vertical profiling float, mrxFLOAT, is constructed with a 4" acrylic tube and operated with a **buoyancy engine of a continuous rotation servo and lead screws**. The mrxFLOAT is 31.4cm in height. The top section of the float holds the electronics, which consist of 4 AA batteries, an inline 7.5A fuse, an on/off switch, a 6V to 3.3V buck converter, and an ESP32 LoRa V3. The actuator is screwed through a nut and can extend 10cm vertically. Another ESP32 LoRa V3 transmits and receives signals, consisting of the local time, ROV team number, and pressure information. The LoRa radio protocol was selected for its long-range, reliability, and ease of use. MUREX considered options like Wi-Fi and Bluetooth but preferred the simplicity and long range of LoRa 915 MHz.



mrxFLOAT. Photo: Y. Shah

Following safety specification ELEC-NDR-005, a 7.5A fuse is attached within 5cm of the power wire of the battery. The enclosure has been verified to **comply with MECH-001** (pressure tested to 5m water column depth). The pressure relief system consists of a top cap which disengages from the acrylic tube in the event of high pressure, which is also paired with a standard blue robotics pressure cap rated to 12 psi. The depth sensor collects data in 5-second intervals throughout the ascension and descension period.

BUILD/BUY, NEW/USED

My approach to choosing the right components; **if I find that a task can be done better than commercial alternatives (higher efficiency for same price, same efficiency for lower price, higher efficiency for lower price, features that cannot be purchased) I design and test in-house**. If industry standards exceed my team's capabilities for a reasonable price, I use the commercial alternative. I make extensive use of custom boards in our electrical system because they greatly reduce the cost and space while increasing performance relative to what I could purchase. Below is a table exemplifying the cost-effectiveness and innovation of our electrical system. The performance and cost of the electrical system allows more budgeting towards other areas of research and development while providing a state-of-the-art control system that can adapt to any deployment environment. The custom parts in my frame design were scrutinized to ensure that the value they contributed justified the added complexity.

My thruster mounts are custom CNC'ed aluminum brackets which far exceed the strength and rigidity of our previous 3D-printed mounts. My 3D-printed parts are produced at a low cost and provide mounting angles that would not be possible with off-the-shelf parts. The CNC-cut HDPE plates used to mount the electrical enclosure, thruster and camera help simplify mounting configurations. They were produced at a low cost, using previously owned material and our school theatre's CNC.

To conserve cost, I have re-used the ROV's controller, thrusters, various miscellaneous hardware, and re-implemented our old ROV electronics enclosure as an effective solution for the mrxFLOAT. The latter in particular saved us over \$300 in cost in comparison to purchasing a new product. In pivoting to our newest ROV design, nearly all aspects of the system have been reconsidered, thus very few parts have been re-used in the MUREX ROV V3.5.

System	Function	Prototype Price (ea) [15]	Commercial Option Price (ea) [16] [17]	Cost Savings
MUREX Ethernet Switch	Packet switching on robot network	\$9	Blue Robotics Ethernet Switch: \$175 BotBloX SwtichbloX: \$72	95%-59%
MUREX Carrier Board	Full system control	\$30	Holybro Pixhawk RPi CM4 Baseboard: \$446.99	93%
MUREX Power Board	Power conditioning and dual DC/DC converter	\$20	Sky-Drones SmartAP PDB: \$90	78%
MUREX ANYESC	Motion control system	\$14 * 8	Flyduino KISS ESC: €22.9, ~\$25 * 8 Holybro Kotleta20: \$57.99 * 8	76%-44%
OFDM Data Injection (mrxPLC)	Data transfer across tether	\$0	Fathom ROV Tether: > \$220 (increases with length)	N/A (Zero Cost)
Total		\$171	\$1,395.91	>88%

CRITICAL ANALYSIS AND TESTING

MUREX's extensive use of custom electrical and software components result in high performance at the expense of complex testing methodologies. **Our testing methodology focuses on variable isolation and finding the root of the problem as quickly as possible.** Our mechanical team utilizes extensive use of iterative design and 3D printing to improve designs. I conduct pool tests as often as possible to maximize iterative design potential.

MUREX mechanical has produced over 5 chassis designs to achieve the *MUREX ROV V3.5 design*. **Each design has been run through extensive fluid simulations in Autodesk CFD** providing the necessary insight to isolate places of excess water resistance that helped us relocate the positions of electrical enclosures, HDPE plates, and thrusters. Each new design tailors more towards the RFP's tasks requirements by adding, and changing placement of mission sensors and manipulators.



V1.8, HDPE Frame

~ excess water resistance for vertical profiling, unadaptable mounting opportunities

V2.0, First Exoskeleton Frame

~ bulky, unstable and weak acrylic base

V3.1, Hybrid Winged Design

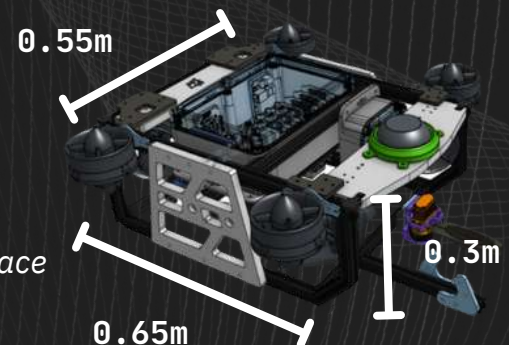
~ unreliable vertical profiling due to water resistance, no manipulator space

V3.2, Compacted Wing Design

~ too small, no carrier claw space

V3.3 + V3.4, Medium Wing Design with Front Bezel

~ Thruster Mounting Unstable



Top left to bottom right: V1.8, V2.0, V3.1, V3.2, V3.3, V3.4, V3.5

SOFTWARE SYSTEM

Before settling on MRX, **a modified codebase upon Ardupilot tailored for our system**, I extensively tested alternative control systems. Our codebase has seen 3 major revisions, each with its own benefits and disadvantages. Using MRX ensures that I make full use of our ROV capabilities.

Control System	System Coverage	Setup Difficulty	Debugging Difficulty
MRX V1: Python	Low (no sensor integration)	Low	Low
MRX V2: Rust	Medium (telemetry collected from most sensors + minimal active monitoring)	High	High
MRX V3: Ardupilot	High (active stabilization + real-time telemetry)	Medium	Medium

By using an off-the-shelf approach for programming I spent our time modifying the codebase, tailoring it to our system rather than writing our own system and fixing bugs as they arise. This way, I ensure that the minimum viable system worked and MUREX spends time optimizing a system that already worked reliably and consistently. **ArduSub fulfills many of our system requirements but falls short of a few important ones:** thruster control, camera integration, and peripheral arm control. The final state of our codebase can be split into the following categories: ArduPilot tooling, Lua Scripting, Component Test Scripts, ANYESC Firmware, MASCP, and Diagnostics Program.

Notes: [10] See [ArduPilot EKF explanation + theory](#).
[11] [Understanding altitude](#).

../ ARDUPILOT TOOLING

ArduSub is the glue that brings together the electrical system. A well-designed control system is the crux of MUREX's granular and reliable driver control. A 22-state Extended Kalman Filter (EKF) accurately estimates the ROV's attitude and heading by **fusing sensor data from our IMU, gyroscope, magnetometer, and depth sensor [10]**. ArudPilot's advanced EKF and Attitude and Heading Reference System (AHRS) allows for active stabilization on all 6 degrees of freedom, ensuring that the ROV only moves when intended [11]. Active stabilization corrects imperfections in the ROV construction allowing for more flexibility in component placement around the ROV. Any imperfections in buoyancy and balance become unnoticeable when stabilization active.

Our driver interface/ground control system is also part of the ArduPilot software suite. MUREX uses a custom version of QGroundControl (QGC) compiled specifically for our topside computer to **aggregate telemetry and communicate instructions** from the driver to the ROV. QGC greatly simplifies the testing and tuning of the system. Our programming and electrical teams can focus on parts of the system that were developed in-house without worrying whether a bug is caused by the systems that control it or the systems that were built to observe it.

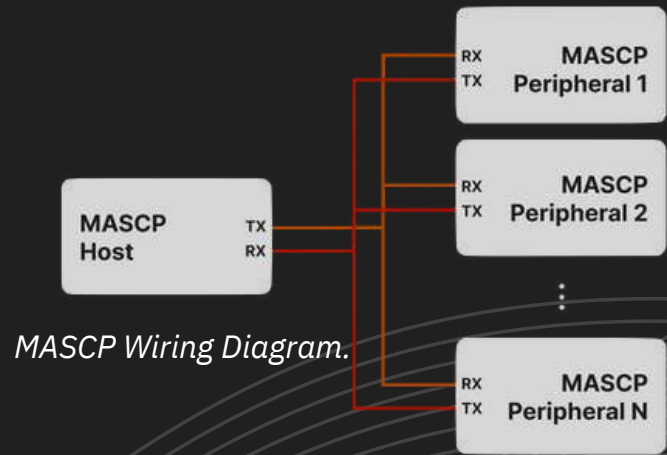


~/ MASCP

A simple standardized communication scheme that could be easily modified in software and implemented in hardware was critical for minimizing complexity in communication within the electrical enclosures. As a result, MUREX Asynchronous Serial Communication Protocol (MASCP) was created. **MASCP is a half-duplex communication protocol capable of enabling communication between any devices connected with only 2 wires and on common ground.**

MASCP is built on top of UART [13], a very simple and reliable serial communication protocol. The data format for MASCP was inspired by I2C. The working principles of MASCP involve the host and several peripheral devices. Every device can communicate with every other device in the network using unique addressing. The host device acts as a transmitter for messages that don't contain the host address. So, I make the most powerful processor in the system the host.

Every device is assigned an 8-bit address which cannot be modified without reflashing the firmware on the device. **Every device must have a unique address to enable individual addressing and effective communication between intended devices.** To communicate any device in the system can send an MASCP payload with the target address at the first byte. The host has logic implemented to distinguish between packets intended for itself and other devices in the network, forwarding any messages as needed. While I originally proposed MASCP as a solution for communication between our motion systems and the carrier board, I now use MASCP for any peripheral that requires external instructions for operation.



MASCP Wiring Diagram.

./ LUA SCRIPTING

I use Lua Scripting with ArduPilot to extend the functionality of ArduSub to the **MUREX ecosystem without increasing its complexity [12]**. ArduPilot exposes a Lua Scripting API that supports reading and modifying the execution of ArduPilot.

/ COMPONENT TESTING SCRIPTS

In addition to the tools provided by ArduPilot, I also have our own set of scripts to ensure that modifications to the electrical system maintain functionality. Most of our test scripts are written in the Rust programming language and are cross-compiled to run on the MUREX Carrier Board. Before switching to ArduPilot, MUREX conducted extensive research on using Rust for embedded and high-level applications. Ultimately, I found that the Rust ecosystem was underdeveloped and too young to be used effectively in our system. Since the scripts worked well and did not need to be re-compiled, I continued to use them to test the system integrity even after switching to ArduSub.

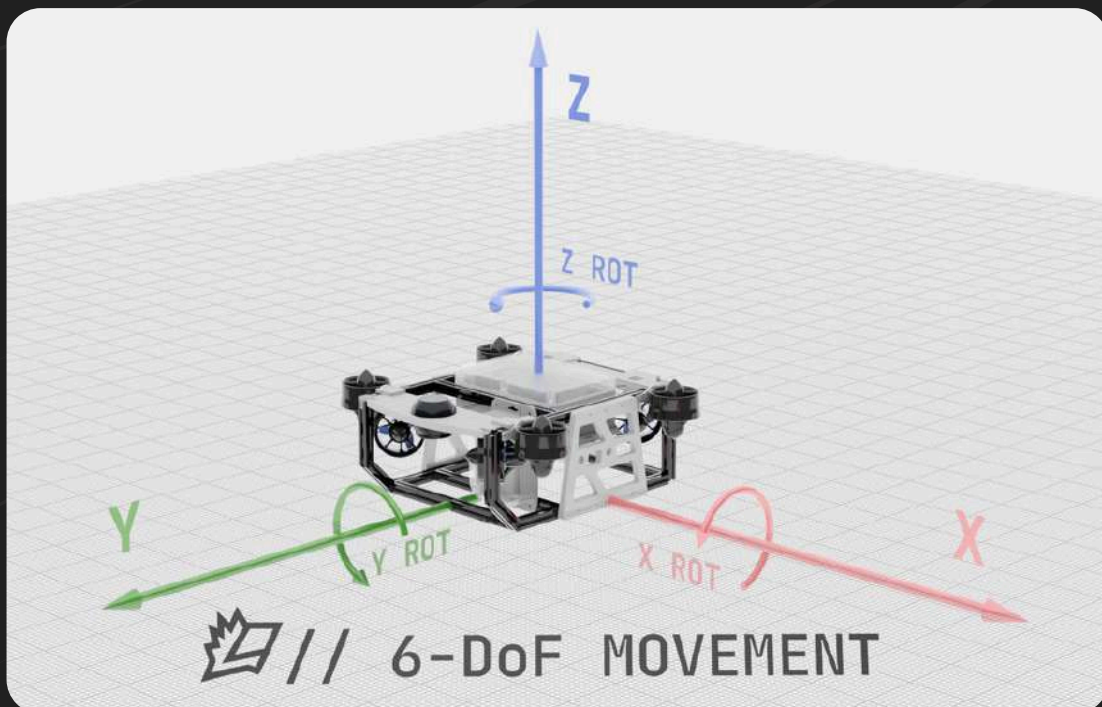
// ANYESC FIRMWARE

The ANYESCs were the first component in the system to integrate MASCP. The core of the ANYESC firmware revolves around initializing the thrusters and listening for MASCP packets for throttle information.

The ANYESC firmware also provides debugging information through Serial lines and status LEDs. I experimented with various embedded approaches like using Rust and MicroPython but finally settled on using C with PlatformIO. PlatformIO is robust and easy to debug allowing for rapid prototyping.

PROPULSION

The MUREX propulsion system consists of 8 Blue Robotics T200 thrusters driven by MUREX ANYESCs. The MUREX ROV has 8 thrusters in a vectored configuration, allowing movement in all 6 DoFs. This is a standard thrust vectoring configuration, allowing for rapid electrical and software expansion. **Using our custom control system framework, mrx, the MUREX ROV's advanced 10-DoF electrical system allows the robot to stabilize and depth-hold** at any location and depth in the deployment environment. Their external position to the ROV, provides a greater torque and stability but increases chances of entanglement with tether, and SMART Cables. This increased chance of entanglement has been solved with the use of reflective tape and thruster guards around all directions off the thrusters.

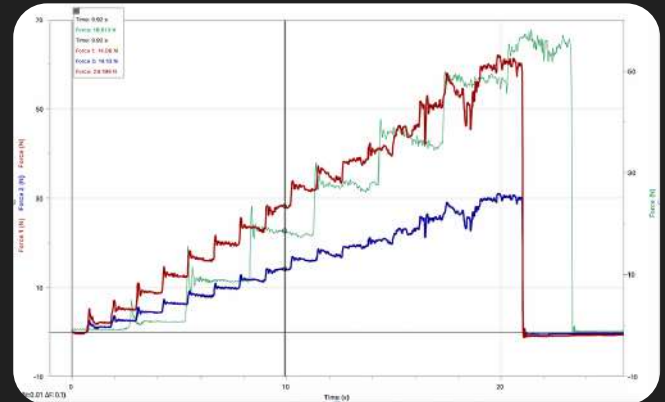


PROPULSION CONT.

I conducted extensive experimentation of the current draw and efficiency of trapezoidal, field-oriented control (FOC), and sinusoidal drive of brushless DC motors. FOC offers clear advantages of higher torque and efficiency at the cost of higher computational requirements. My schedule didn't allow for further experimentation of FOC driving of BLDCs, so I continued our use of trapezoidal drive ESCs running a custom version of the BLHeli_S firmware. Mounted in each ANYESC is a XILO 25A ESC, chosen for its very high performance-to-cost ratio.

BLDC Control Method Comparison [14]:

Control Method	Trapezoidal	Field-Oriented Control	Sinusoidal
Software Complexity and Motor Efficiency	Low	High	Medium
Maximum Speed	High	High	Medium
Noise	High	Low	Low
MOSFET Switching Losses	Low	High	High
Final Choice	Selected	Under development	Unsuitable



FOC and Trapezoidal Thruster Profiling.

BUOYANCY AND BALLAST - MATH & PHYSICS CALCULATIONS

The MUREX ROV was specifically designed with buoyancy and ballasts in mind. The mechanical design used CAD software such as Fusion 360 to determine the center of gravity and the density of the robot. The use of Fusion 360's in-built dimensional analysis helped approximate placements of all ROV elements to maintain a neutrally buoyant and stable ROV. With the addition of well-placed foam pool noodles, the **MUREX ROV V3.5 has a specific gravity of 1.002**. The most buoyant part of the system, the Main Enclosure, is placed at the center and above all other components to heighten the center of mass which aids in the stability and safety in case of unwanted rams of the outermost ROV sections such as the thruster guards against oceanic facets.

The combination of recycled HDPE and aluminum extrusions allows the chassis to be nearly neutrally buoyant. This is due to HDPE's density of around 0.97 g/cm^3 . The aluminum's density (2.7 g/cm^3) compensates for the electronics' density ($\sim 0.8 \text{ g/cm}^3$). The decision to use 2020 aluminum allowed for easy mounting and testing of various ballasts and buoyancy materials. After further CAD and physical corrections, the MUREX ROV is **almost exactly neutrally buoyant**. In combination with the control system, the MUREX ROV can hover at any position when completing tasks which is an immense aid to the driver who wishes to complete tasks designated in the RFP requiring the most accuracy such as

The initial mass of the mrxFLOAT rests at 3407 g, but once 100g of water is intaked through the syringes, the mass increases to 3507g while the volume of the float stays constant at 3463 cm^3 . Thus, the intake of water changes density from 0.984 to 1.02 resulting in descension. The inverse is repeated for ascension.

PAYLOAD & TOOLS

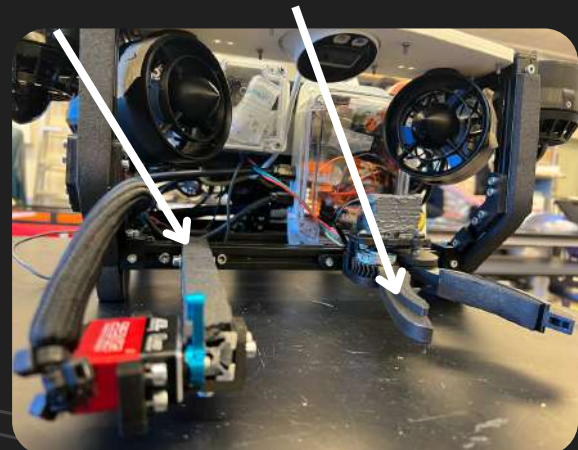
There are a total of four cameras on the ROV, each providing the ROV with a deliberate, unique view. The downward angled camera, mounted at a 45 degree angle on the chassis' front HDPE crossbeam is an Amcrest IP5M-T1277EW-AI. It provides a frontal and slightly downward view of the claw and its surroundings. There is one Logitech Brio 101 camera and one Logitech C270 camera, one facing the front in a mini enclosure (ML-31F) and one facing upwards in the main enclosure. This watertight placement allowed us to remove the outer covers of both cameras and save space. The Brio 101 camera inside the mini enclosure provides a front view of the claw and looks directly ahead. The C270 camera, facing upwards in the main enclosure, helps the driver position the ROV in the deployment area as it returns to the surface while also giving the driver a rough idea of ROV's depth.

The last camera, angled 45 degrees downwards in the back of the ROV, is an Amcrest IP5M-T1179EB-28MM. While at the back of the ROV, this camera faces the front and gives the driver an overview of the pool floor and a wider view of the ROV's surroundings. This camera is used for creating an photogrammetric model of oceanic ecosystems and corals. The camera mount, while sturdy, allows movement of camera so as to fit the driver's needs. These four cameras allows the driver to operate the ROV, track its movements, and approximate its location remotely. The front facing Logitech and the downwards facing Amcrest focus on the two claws and target various objects, while the other two cameras provide upward and downward facing angles to give the driver an approximate idea of the ROV depth.

Left to Right: Logitech Brio 101 in Mini Enclosure. Front Amcrest. Back Amcrest.



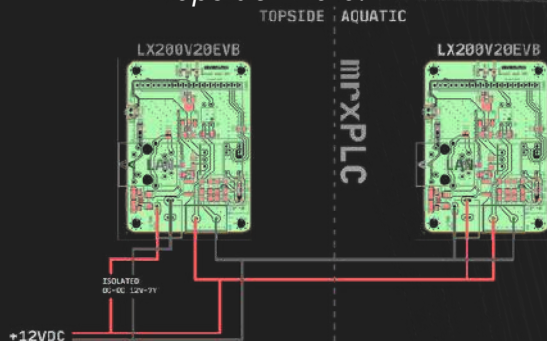
The two manipulators, denoted as active and carrier claw, on the front of the ROV perform all the tasks outlined in the MATE RFP. **The active arm is a two-part geared claw mechanism that extends out of the ROV for easy access into remote areas.** This claw is used to perform smaller tasks requiring more accuracy including the relocation of ocean observing assets, retrieving and installing power connectors, and transplanting the brain and branching corals. The main innovation within the active arm includes the custom waterproof servo, and the herringbone gears. The oil-filled and waterproof enclosed servo in the active arm is positioned downwards to use gravity to prevent water from entering and damaging the servo. The use of custom designed herringbone gears in contrast to traditional gears gives the active-arm an increased grip and reduced chance of wear and tear. The carrier claw, located in the bottom, provides a far sturdier base to transport heavier freight including the deployment of the probiotic irrigation system.



Driver - Interface



Topside Photo.



mrxPLC Functional Diagram.

TETHER DESIGN RATIONALE

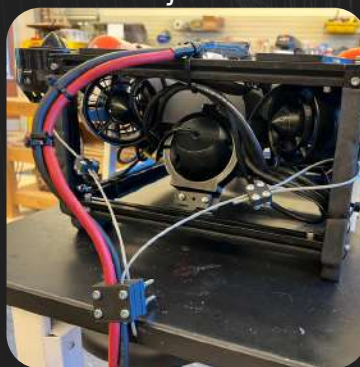
The strain relief system on the back of the ROV uses metal wire as a medium **to take the 99.5% of the physical strain off the main tether**. The tether wires are friction clamped to two metal wire loops, connecting to the structural frame of the ROV in the back. MUREX conducted multiple iterations of strain relieving, ultimately settling on this design for its simplicity and reliability. The distributed stress from using two metal wires allows the strain relief to be so robust that it supports the weight of the robot for extended periods of time.

The tether is highly flexible and experimentally modified with flotations to be neutrally buoyant. Tether managers are trained in communication and deployment to avoid possible entanglement. In addition, tape around the thrusters creates an imaginary box around the ROV disallowing entanglement with tether, SMART cables, and recovery lines.

TOPSIDE SYSTEM

The MUREX Topside is built around an Amazon Basics Suitcase that I've heavily modified. A 23.8" monitor is attached to one side, while the other has DC and AC wiring. The demo team has access to all topside wiring through an easily removable PETG top layer. The suitcase has ample space to hold all the required electronics as well as tools and the controller. All connections into and out of the topside are strain-relieved. The Anderson Powerpole input is fused on the high side and passed through the emergency stop button and power monitor in series. **The main DC input is passed to the robot through the tether connection and the mrxPLC system in parallel [1].** In particular, the power tether is secured to the topside with two industrial locking connectors, rated to support 30A within temperature regulations. The topside uses USB-C to enable USB-C charging, USB device connection, and HDMI output to the monitor. A suitcase was selected for its low price, wide availability, and thin shell which allowed for modifications to be made. Costly pelican cases, unreliable DIY options, and unprotective enclosure-less/exposed options were also considered, but the commercial suitcase struck a fair middle ground. Upon completing a deployment, the demo team can be ready for departure efficiently by pressing the stop button, detaching from AC power and DC power, and zipping up the suitcase. The wheels on the suitcase also allow the topside to be transported with ease.

Strain Relief. Photo: O. Chang



MISSION SENSORS

MUREX dedicated countless R&D hours to developing accurate and effective mission sensors. This includes a high-accuracy (0.5C deviation) **temperature sensor** communicating over the Dallas One-Wire protocol to the Carrier Board that is checked with the smart cable sensor readings and a **hall-effect rotation sensor** on each manipulator for sub degree level rotational accuracy on ADCP probes, retrieving acoustic receivers and transplanting coral.



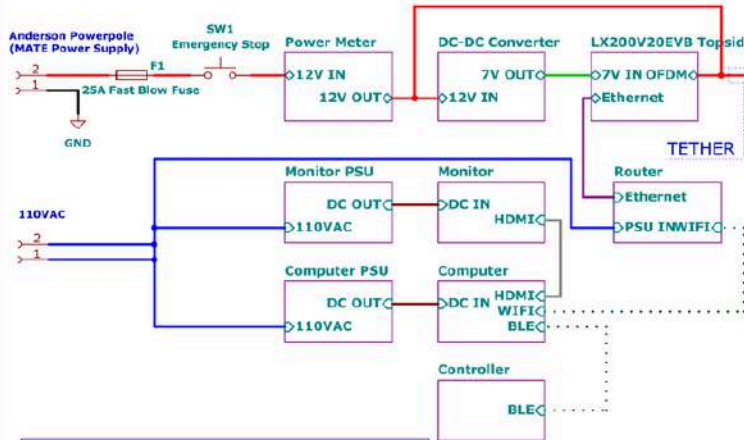
MUREX ROBOTICS

ATTEMPT THE IMPOSSIBLE

SYSTEM INTEGRATION DIAGRAMS

SYSTEM INTEGRATION DIAGRAM (SID)

* FLUID POWER IS NOT USED



ELEC-008R Requirements

8 thrusters	2.5A each = 20A
3 cameras	0.2A each = 0.6A
8 ESCs	0.05A each = 0.4A
1 ethernet switch	0.1A = 0.1A
1 control board	0.2A = 0.2A
4 servos	0.8A each = 3.2A
2 PLC modules	0.1A each = 0.2A

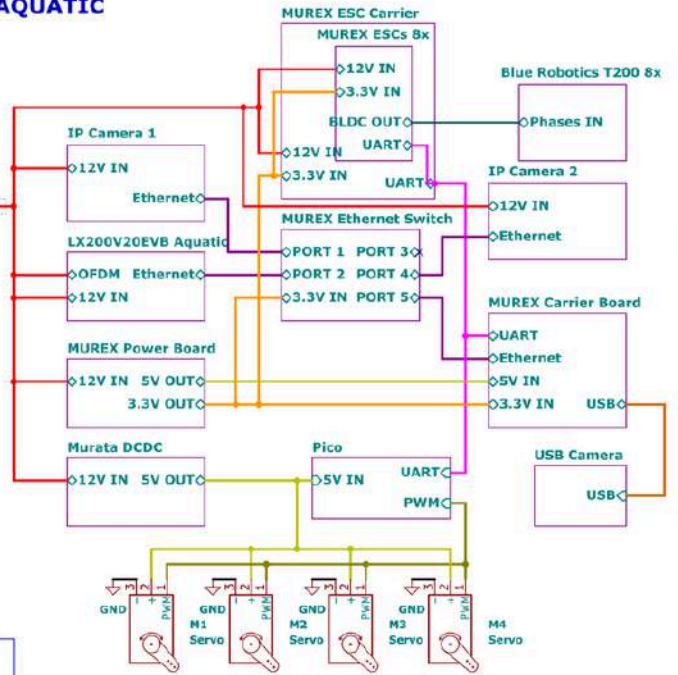
Total Amps 24.7A * 150% = 37.1A

Fuse Selected: 25A ATO

SID Legend

110VAC	5VDC
PSUs DC	GND
12VDC	UART (MASCIP)
Ethernet	BLDC Phases
3.3VDC	Wireless
PWM	7VDC
USB	

AQUATIC



MUREX ROY V3
SYSTEM INTEGRATION DIAGRAM (SID)

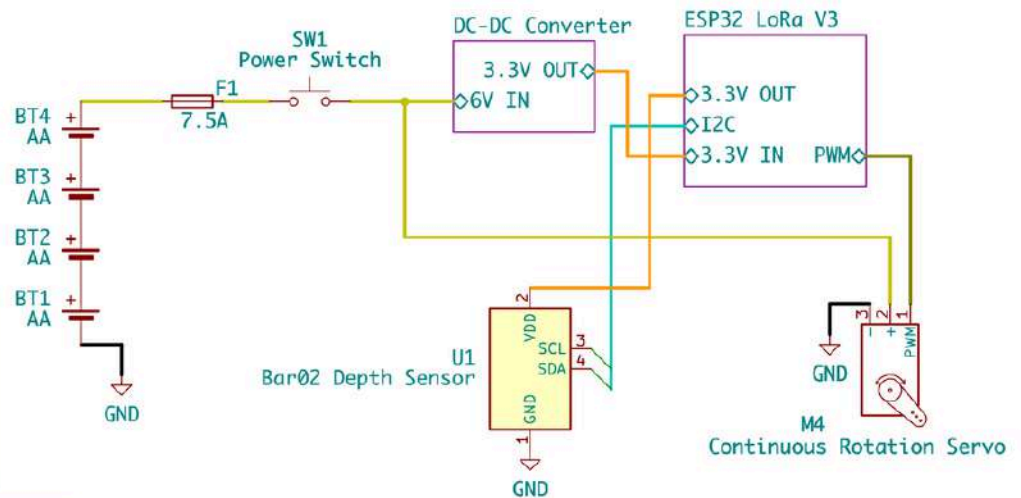
MUREX ROBOTICS
Sheet: /
File: SID.kicad_sch

Title: System Integration Diagram

Size: User Date: 2024-05-19
KiCad E.D.A. 0.99.0-1133-q4f3117aes0

Rev: V3.5
Id: 1/22

mrxFLOAT SID



SID Legend

3.3VDC	6VDC
PWM	GND
I2C	

ELEC-008R Requirements

CR Servo	4A = 4A
DC-DC Parasitic Power	0.3A = 0.3A
ESP32 LoRa V3	0.5A = 0.5A

Total Amps 4.8A * 150% = 7.2A

Fuse Selected: 7.5A ATO

mrxFLOAT V9
FLOAT SID

MUREX ROBOTICS
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File: float_sid.kicad_sch



MUREX ROBOTICS

ATTEMPT THE IMPOSSIBLE

THANK YOU TO:

- **MATE II** for hosting the MATE ROV Competition
- **MATE ROV World Championships** for hosting this competition
- **MATE ROV NE Regionals** for invaluable advice and providing this opportunity
- **The Phillips Exeter Academy Theater Department** for providing CNC machining
- **JLCPCB** for printed circuit board production
- **PTC OnShape** for CAD
- **KiCAD** for electronics design software
- **Bambu Lab and Prusa Research** for providing 3D printers

ACKNOWLEDGEMENTS AND REFERENCES

PURCHASES / VENDORS

- *AGFRC and Savox for supplying servos.*
- *McMaster-Carr, Amazon.com, Digikey, Mouser, Polycase, Fabworks, Misumi, Lowes, Blue Robotics, Adafruit, GoBilda, GetFPV, and others for supplying parts.*



ONLINE RESOURCES

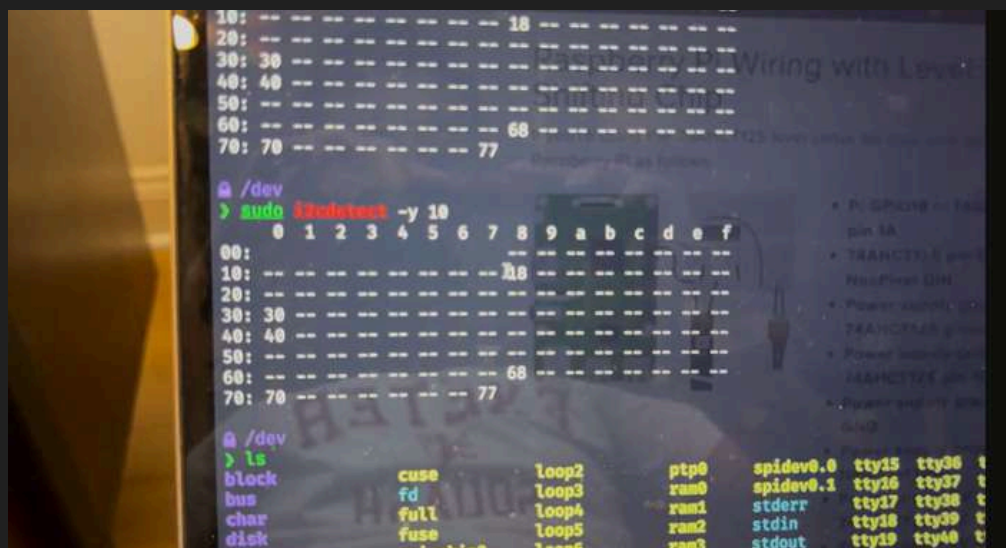
- *TI, STMicro, IC Plus, Bosch Sensortec, TE Connectivity, Diodes Incorporated, Winbond, Molex, Qualcomm, Broadcom, Raspberry Pi Foundation, Rockchip, Allwinner, Logitech, TNK, Murata, Panasonic, WAGO, Würth Elektronik and others for providing online resources and whitepapers. My system either includes/uses electrical components designed by the above companies or has read whitepaper publications. **However, I did not receive advice from any company listed above.***

SINCERE GRATITUDE TO:

- **Phillips Exeter Academy** for providing space, funding, and support to the MUREX Robotics Company.
- **John H. Hessel '52 and Sidney A. Hessel** for providing funding as part of the Phillips Exeter Academy Hessel Fund.

[illegible]

Carrier Board V1 testing with HDMI



Carrier Board V1 I2C Debugging



NeoPixel Testing

V1 --> V3

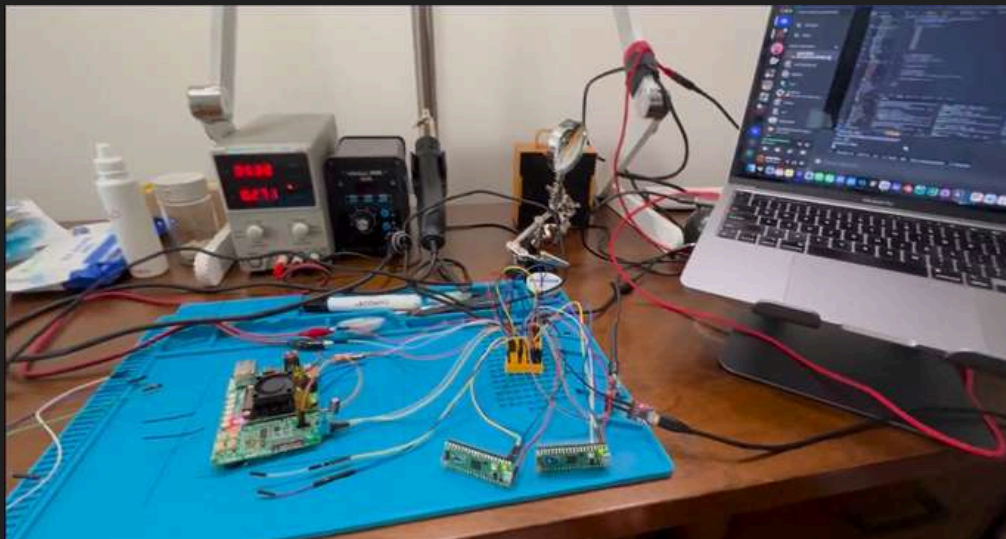


42% Area Decrease

Other Testing



ESP 32 Flashing



MASCP Demo

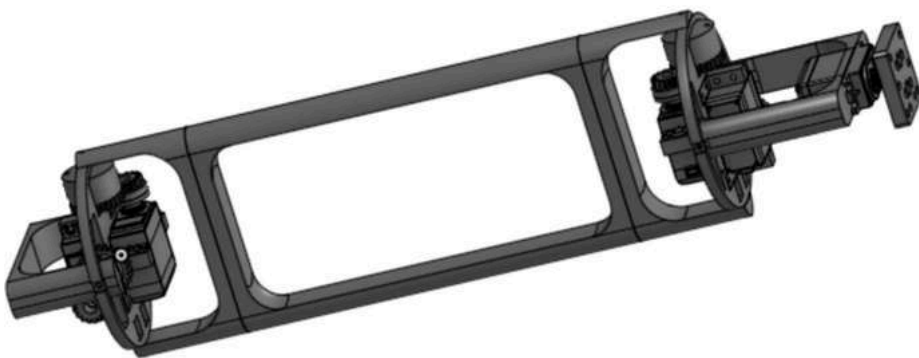
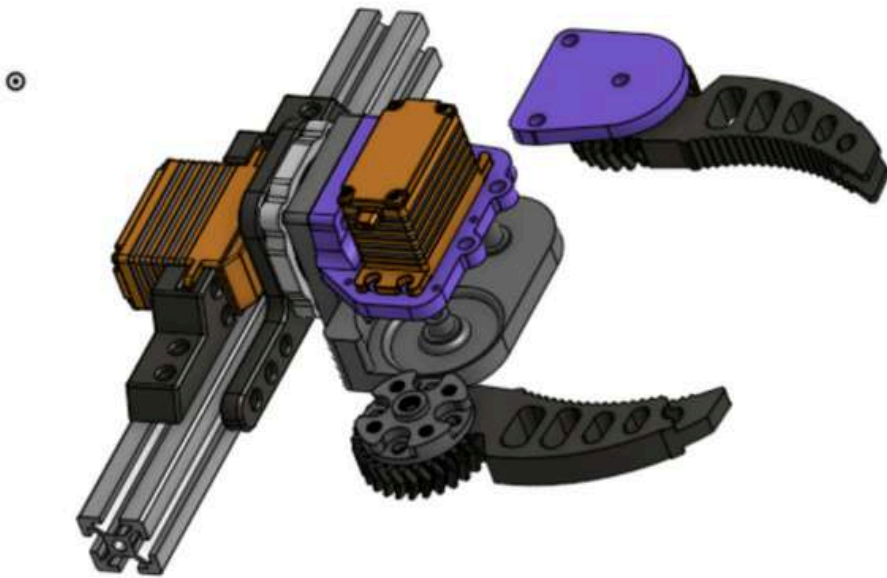


ArduPilot-MASCP
Integration Working
FINALLY!

MUREX ARMS



Waterproof ServoCasing



BME680
Low power gas, pressure, temperature & humidity sensor

The BME680 is a digital 4-in-1 sensor with gas, humidity, pressure and temperature sensing principles. The sensor module is housed in an extremely compact footprint of only 3.0 x 3.0 mm² with a maximum height of 1.00 mm (0.93 ± 0.07 mm).

- Typical applications
• Indoor air quality
• Home automation and control
• Internet of things
• Weather forecast
• GPS enhancement (e.g. time-to-first-fix improvement, dead reckoning, slop)
• Indoor navigation (change of floor detection, elevator detection)
• Outdoor navigation, leisure and sports applications
• Vertical velocity indication (rise/sink speed)

- Target Devices
• Handsets such as mobile phones, tablet PCs, GPS devices
• Wearables
• Home weather stations
• Smart watches
• Home appliances
• Navigation systems



SIRA00DP
Vishay Siliconix

N-Channel 30 V (D-S) MOSFET



- FEATURES
• TrenchFET® Gen IV power MOSFET
• 100 % RDS and UIS tested
• Material categorization: for definitions of compliance please see www.vishay.com/doc79912
• APPLICATIONS
• Synchronous rectification
• OR-ing
• High power density DC/DC
• VRMs and embedded DC/DC

Table with 2 columns: Parameter, Value. Rows include VDS, Power max, RDS(on), Qg, IRL, and Configuration.

Table with 4 columns: Parameter, Symbol, Limit, Unit. Rows include Absolute Maximum Ratings (VDS, VGS, ID, ISM) and Ordering Information.

- Description
Features
Applications
• µPD720202 is a USB host controller LSI compatible with the USB 3.0 and xHCI (eXtensible Host Controller Interface) 1.0 specifications.
• µPD720202 comes in a 48-pin QFN package and has the following improvements over previous products.
• Reduced power consumption in low-power mode to 4.5mW, a reduction of 90% compared with Renesas Electronics' existing products.
• Improvements to the data transfer processing circuit provide an increase in the effective data write speed of more than 40% from existing products.
• Firmware download function eliminates the need for external serial flash ROM.

Product details

Table with 4 columns: Parameters, Package | Pins | Size, Features, Description. Rows include Rating, Operating temperature range, Topology, Type, Iout, Vin, Switching frequency, and Switching frequency (max).



TPS25947xx, 2.7-V-23-V, 5.5-A, 28-mΩ True Reverse Current Blocking eFuse with Input Reverse Polarity Protection

1 Features

- Wide operating input voltage range: 2.7 V to 23 V
- 28-V absolute maximum
- Withstands negative voltages up to -15 V
• Integrated back-to-back FETs with low on-resistance: RON = 28.3 mΩ (typ.)
• Ideal diode operation with true reverse current blocking
• Fast overvoltage protection
- Overvoltage clamp (OVC) with pin-selectable threshold (3.8 V, 5.7 V, 13.8 V) and 5-µs (typ.) response time OR
- Adjustable overvoltage lockout (OVLO) with 1.2-µs (typ.) response time
• Overcurrent protection with load current monitor output (ILM)
- Active current limit OR circuit-breaker options

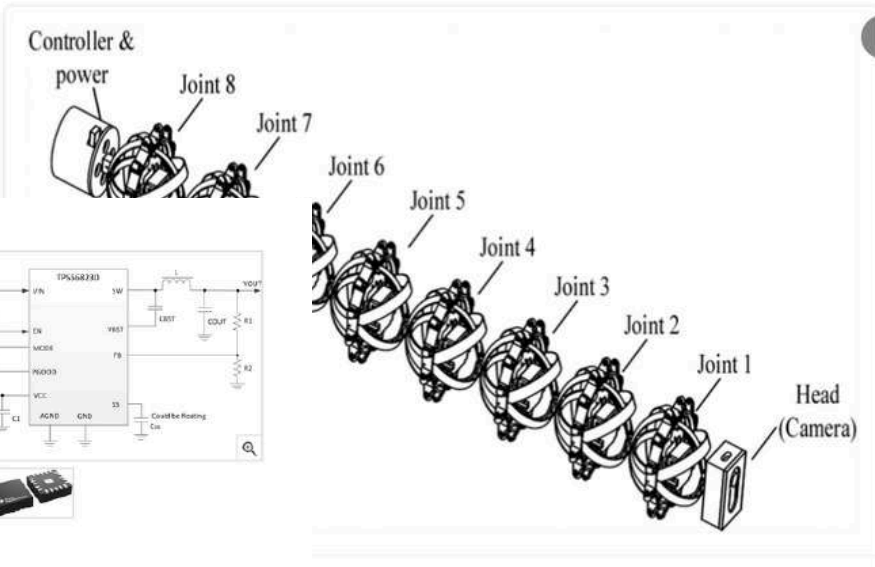
3 Description

The TPS25947xx family of eFuses is a highly integrated circuit protection and power management solution in a small package. The devices provide multiple protection modes using very few external components and are a robust defense against overloads, short-circuits, voltage surges, reverse polarity and excessive inrush current. With integrated back-to-back FETs, reverse current flow from output to input is blocked at all times, making the devices well suited for power MUX/ORing applications as well as systems which need load side energy hold up storage in case input power supply fails. The devices use linear ORing based scheme to ensure almost zero DC reverse current and emulate ideal diode behavior with minimum forward voltage drop and power dissipation.

Output slew rate and inrush current can be adjusted using a single external capacitor. Loads are protected from input overvoltage conditions either by clamping the output to a safe fixed maximum voltage (pin selectable), or by cutting off the output if input exceeds an adjustable overvoltage threshold. The devices respond to output overload by actively limiting the current or breaking the circuit. The output current limit threshold as well as the transient overcurrent blanking timer are user adjustable. The current limit control pin also functions as an analog load current monitor.

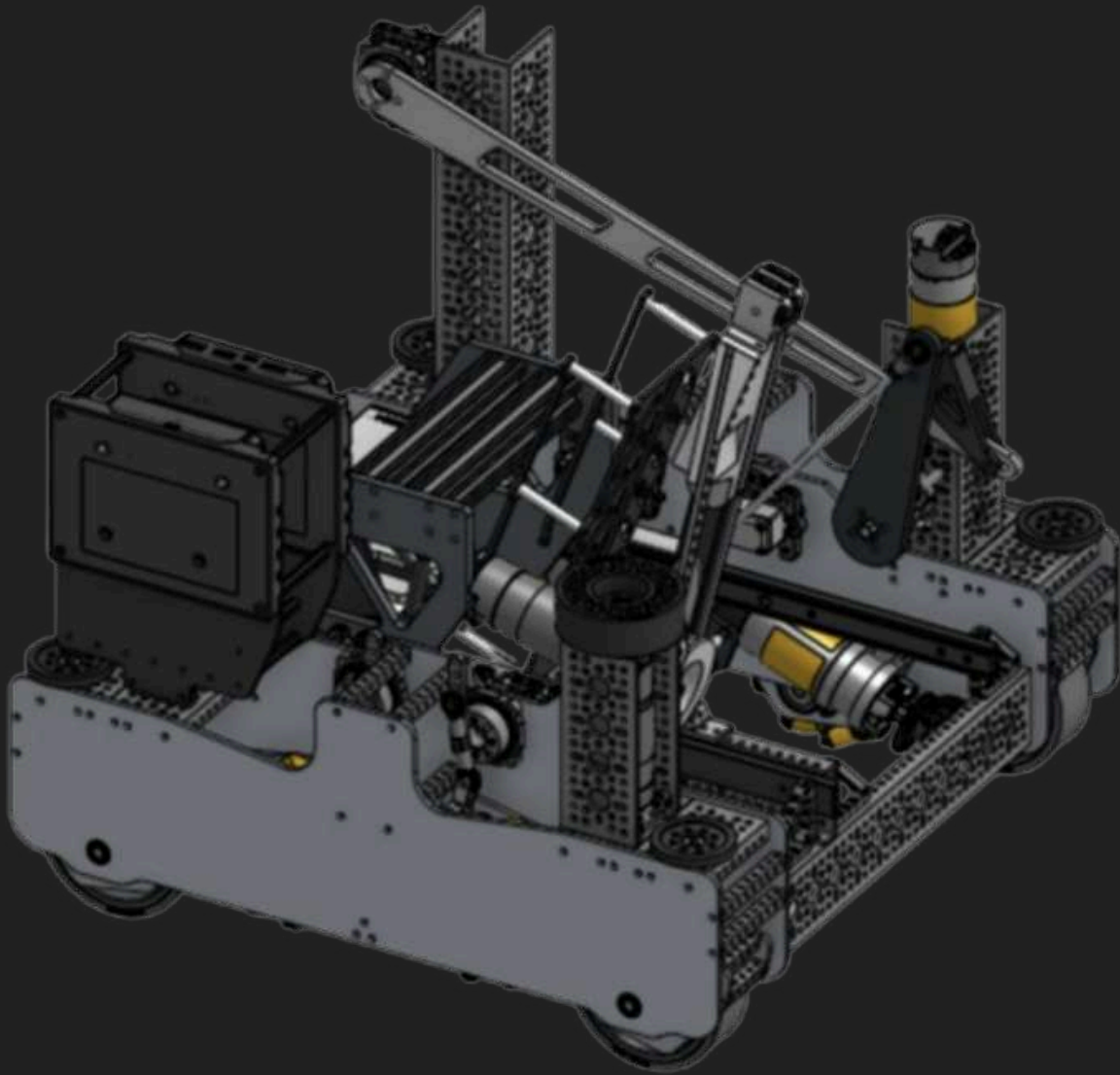
- SPI, LCD interface, Camera interface, UART, I2C, I2S, remote control, pulse counter, LED PWM, full-speed USB 2.0 OTG, USB Serial/JTAG controller, MCPWM, SDIO host controller, GDMA, TWAI® controller (compatible with ISO 11898-1, i.e. CAN Specification 2.0), ADC, touch sensor, temperature sensor, timers and watchdogs

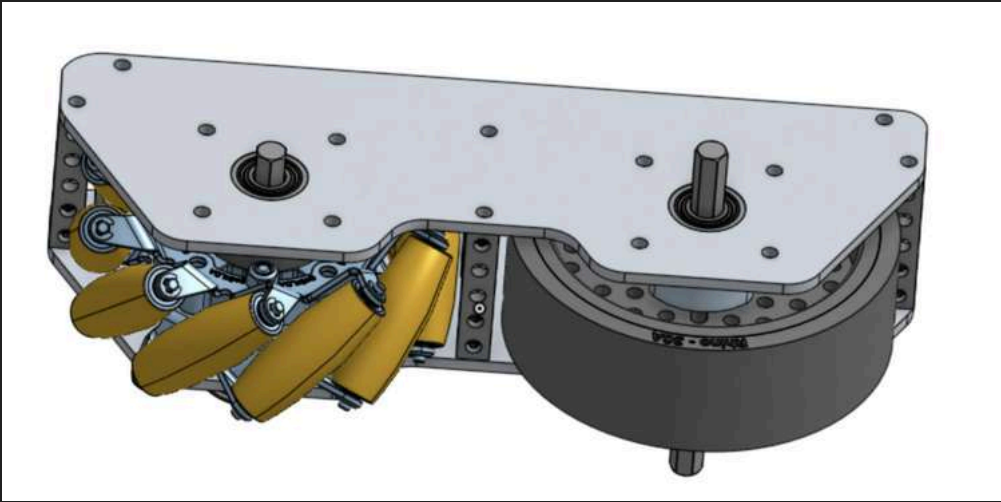
Figure 4 is the 3D drawing of the assembled HITSZ-Snakebot II. In order to reduce the friction with ground, we add passive wheels around the joint. The parameters of the articulation mechanism are listed in Table 3.



FTC Seasons

Team VERTEX (Freight Frenzy)

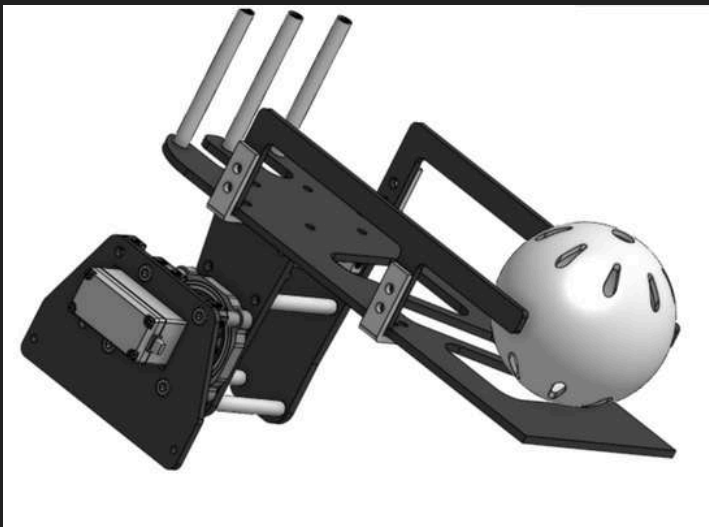




Drive Pod capable of switching between movement flexible mecanum wheel and heavy duty traction wheel.

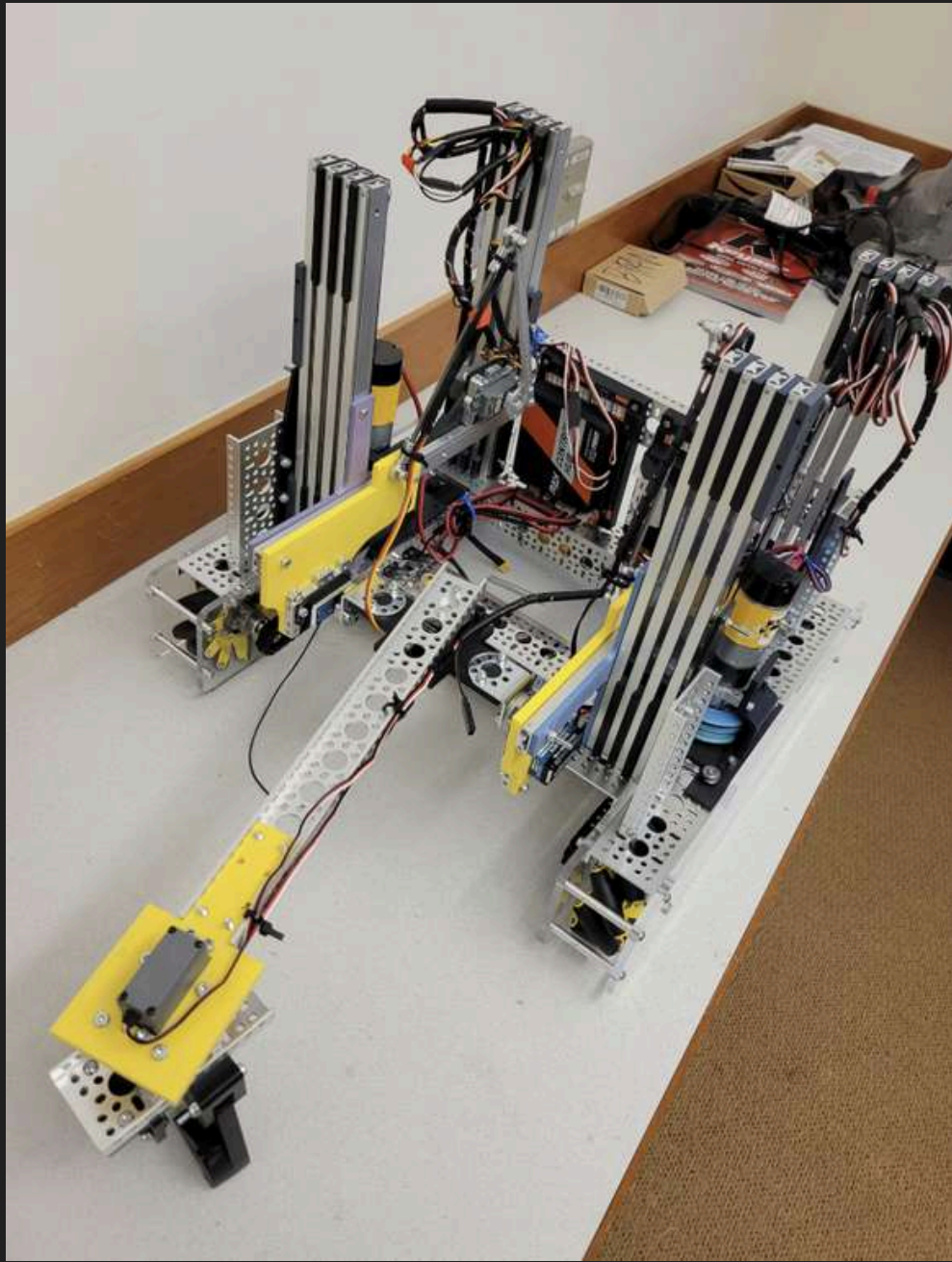


Motor-driven retractable intake system



Swerveable Outtake System

Team APEX (Into the Deep, 12th)



5Dof Arm Capable of:

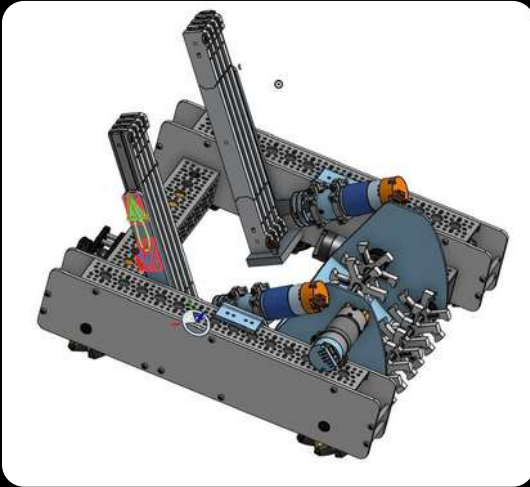
Swivelling and Rotating (Similar to Human Shoulder Ball Joint)

Retraction (Similar to Elbow retraction)

TEAM APEX ROBOT TIMELINE

9

1ST DESIGN: OCT. 2023



Engineering Solution :

- With the centerstage challenge, I chose linear slides, intake, and pinching mechanisms to score pixels at varying heights, aiming for adaptability and precision.

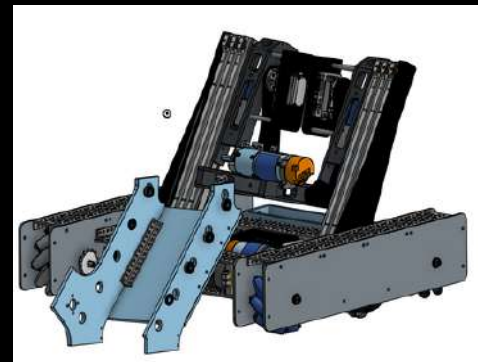
Problem Identification:

- Identified fatal flaws during testing, including issues with motor mounting and slide stringing, through a process of trial and error.

Used Design Process:

- Responded to identified flaws by returning to the drawing board, highlighting the team's commitment to iterative improvement and problem-solving.

QUALIFIER 1: 12/09



Redesign Overview:

- I engaged in extensive redesign efforts to refine and enhance the initial design concept.

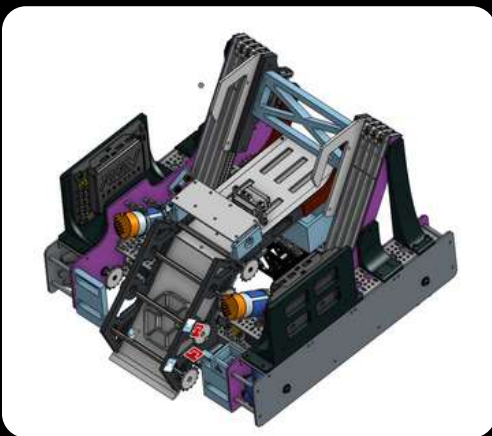
Key Enhancements:

- I implemented improvements including a reinforced slide mount, guided slides, and a customized intake mechanism to increase the robot functionality and performance.

Team Apex's Skill Development :

- Despite limited CAD experience among team members, successfully assembled a functional product, reflecting significant skill development and team achievement throughout the process.

QUALIFIER 2: 01/15



Learning from my First Qualifier:

- Conducted a comprehensive team meeting to share observations from other teams and brainstorm new ideas.
- Adjusted goals within time constraints, prioritizing enhancements in select parts over perfecting the entire robot.

Structural Enhancements:

- Incorporated acrylic plates to reinforce the bot's structure, optimizing internal space for improved versatility and strengthened slides in response to breakage issues.

Integration of 3D Printing Technology:

- Utilized 3D printing technology to model key components such as hubs, battery mounts, and slides within the CAD framework, enabling precise prototyping and customization.

Improvements in Functionality :

- Implemented a stronger and simplified hanging mechanism, along with marginal weight savings in the outtake system and slides, reflecting a commitment to continuous improvement and innovation.



Focused Improvements:

- Directed changes towards improving existing features rather than fixing the entire robot, aiming for efficiency.

Intake Enhancement:

- Modified the intake by adjusting the baseplate angle and design to improve pixel acquisition and ensure smoother axle rotation, minimizing collisions with the plate.

Slide and Pixel Scoring System Enhancement:

- Addressed instability and reliability issues encountered in iteration 3 by redesigning both the slides and the Pixel Scoring System (PSS), focusing on stabilizing the mechanism and enhancing performance.

OVERALL LESSONS LEARNED



1. Acknowledged the importance of setting realistic and achievable goals, enabling better focus and more attainable outcomes.
2. Learned to strike a balance between stability and flexibility in design and implementation, ensuring resilience while maintaining adaptability to evolving circumstances.
3. Embraced the value of learning from mistakes, recognizing them as opportunities for growth and improvement, thereby fostering a culture of continuous learning and refinement.
4. Recognized the role of creativity in problem-solving, leveraging innovative approaches and out-of-the-box thinking to overcome challenges



This being my team's first year competing, I absolutely LOVE what I do. For me, this robot is more than just a bunch of pieces – it's a team effort where every part matters. I take my time building each piece with care, and when I finish one, it's a big win. It's all about the joy of creating something amazing together!

Due to frequent design changes, I often found ourselves lacking the originally planned parts, prompting us to adapt and improvise along the way.

TELE-OP DRIVE TRAIN:

Between the last qualifier and the current one, significant revisions have been made to my code across all subsystems, with particular emphasis on the driving control mechanism

- Addressed issues like drift and inaccuracies by adopting field-centric driving strategy.
- Implemented Road Runner system, utilizing Odometry Pods to track my robot's location accurately.

```
while (opModeIsActive() && !isStopRequested()) {  
    drive.update();  
    Vector2d input = new Vector2d(  
        -gamepad1.left_stick_y,  
        -gamepad1.left_stick_x  
    ).rotated(-poseEstimate.getHeading());  
  
    drive.setWeightedDrivePower(  
        new Pose2d(  
            input.getX(),  
            input.getY(),  
            -gamepad1.right_stick_x  
        )  
    );  
}
```

Vector-Based Driving System:

1. Utilizes vector-based driving, where the robot sets vectors and moves accordingly, enhancing accuracy and precision.
2. Offers an increased reliability compared to previous systems by continuously relying on sensors for accurate positioning.
3. Uses sensor-dependent navigation, ensuring the robot's precise awareness of its position at all times.

my field-centric driving system offers seamless transitioning between Autonomous and TeleOp periods by leveraging localization features to maintain precise position tracking throughout TeleOp

Coordination of Field Coordinates:

- Ensures continuity of field coordinates from Autonomous to TeleOp, optimizing driving accuracy and precision.

Automation of TeleOp Functions:

- Automates specific TeleOp functions, such as blocker opening during outtake lowering, reducing the potential for human error.

Enhanced Efficiency and Reliability:

- Enhances operational efficiency and reliability by integrating automated processes and maintaining consistent positional awareness across periods.

TELE-OP INTAKE AND OUTTAKE:

In Tele-Op, my intake operates in two states: intaking and outputting pixels. Employing switch cases, my intake code manages these stages and tracks the current intake state. Similarly, my outtake in Tele-Op comprises fmy states, also managed by switch cases in the codebase.

```
if (gamepad2.cross) { //if the cross button is pushed, outtake readies for movement
    outtake.move(constants.outtake.READY);
}
if (gamepad2.square) { //if the square button is pushed, outtake moves to the position during movement
    outtake.move(constants.outtake.MOVING);
}
```

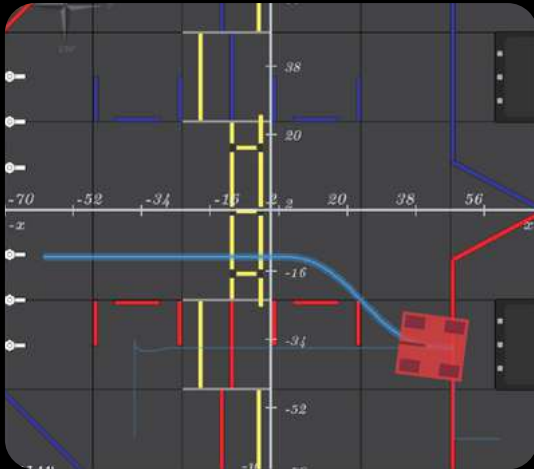
SLIDES:

my slide boasts five predefined heights, supplemented by the flexibility to incorporate a custom height when none of the presets align with a specific target. I developed a function to dynamically set the slide height based on the target, granting my slide full range of motion to score pixels from any height effectively. Additionally, these slides enable my robot to hang on the bars, showcasing their versatility and utility in various competition scenarios.

```
if (gamepad2.dpad_left) { //the left dpad moves the slides to position 1
    slides.preset(constants.slides.FIRST);
}
if (gamepad2.dpad_right) { //the right dpad moves the slides to position 2
    slides.preset(constants.slides.SECOND);
}
if (gamepad2.right_stick_y > 0.2) { //this code allows us to manually control the slide position
    slides.preset(constants.slides.CUSTOM);
    slides.setHeight(slides.getCurrentHeight() + 50); // Switch value as needed
    telemetry.addData( caption: "Custom Slide Height: ", slides.getCurrentHeight());
}
```


PROGRAMMING AUTONOMOUS

13



Autonomous Program Features:

- Based on MeepMeep trajectories, facilitating precise navigation through fmy zones.
- Coordinates the positioning of yellow and purple pixels during the autonomous phase.
- Parks the robot near the backstage area to optimize scoring potential.

Preloaded Pixel Setup:

- Utilizes the OpenCV function to identify the position of the white pixel or team prop, ensuring accurate placement of the purple prop nearby.

Return to Backstage Maneuver:

- Most orientations allow for easy return to the backstage area.
- In challenging positions, an angled spline maneuver aids in accurately positioning the robot for optimal performance.

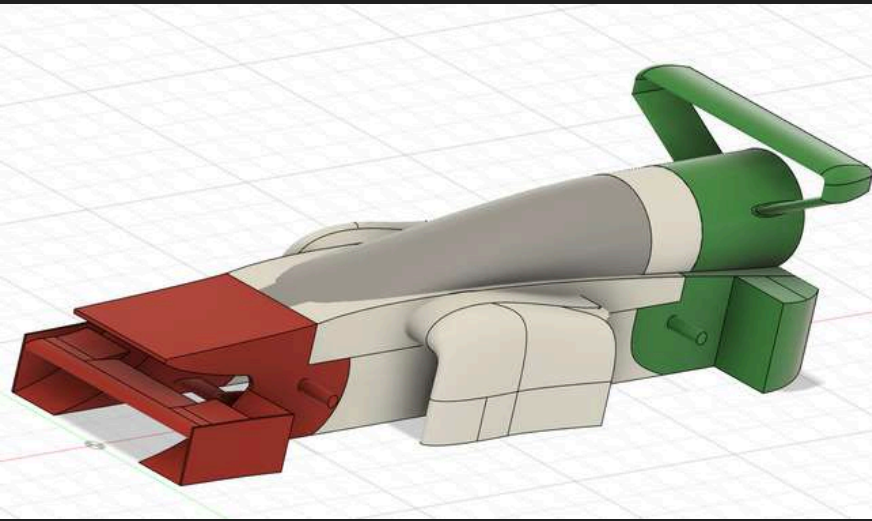
```
yellowPurpleRed.java x
111         .build();
112     }
113
114     TrajectorySequence PARK_LEFT = drive.trajectorySequenceBuilder(SCORE_YELLOW.end())
115         .lineToConstantHeading(new Vector2d( x: 48, y: -12))
116         .lineToConstantHeading(new Vector2d( x: 68, y: -68))
117         .build();
118
119     TrajectorySequence PARK_RIGHT = drive.trajectorySequenceBuilder(SCORE_YELLOW.end())
120         .lineToConstantHeading(new Vector2d( x: 48, y: -68))
121         .lineToConstantHeading(new Vector2d( x: 68, y: -68))
122         .build();
123
124     while (!isStopRequested() && opModeIsActive()) {
125         telemetry.addData( caption: "Zone: ", zone);
126         telemetry.addLine( lineCaption: "");
127         telemetry.addData( caption: "Robot Position: ", drive.getPoseEstimate());
128         telemetry.addLine( lineCaption: "");
129         telemetry.addData( caption: "Current Trajectory: ", current);
130         telemetry.addData( caption: "Next Trajectory: ", next);
131     }
```

After depositing the purple pixel, I retrieve the yellow pixel and return it to the backstage before proceeding to the pixel stacks. I repeat this process twice, intaking two white pixels each time to deposit them on the backstage. This optimizes my scoring strategy and ensures efficient pixel management during the competition

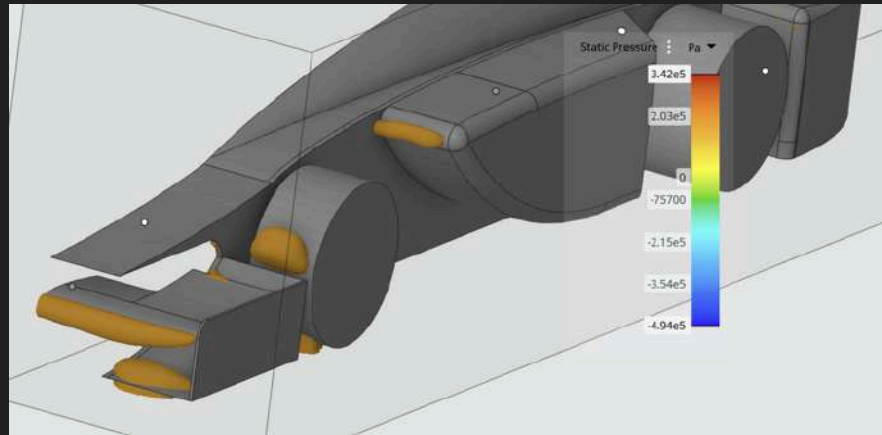
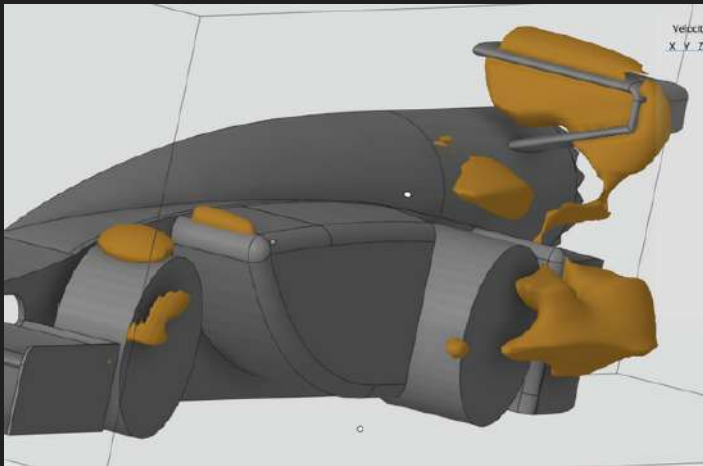
Finally, after parking the robot in its designated space, I easily transition the Autonomous Position to the Field-Centric Tele-ops mode. As an added failsafe measure, I have additional programs that enable parking without pixel intake, minimizing potential disruptions during the competition



F1-in-Schools Competition (TEAM Edge.exe)

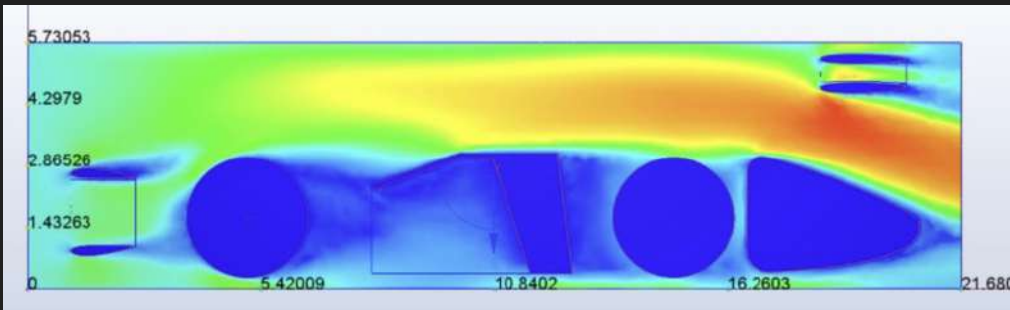
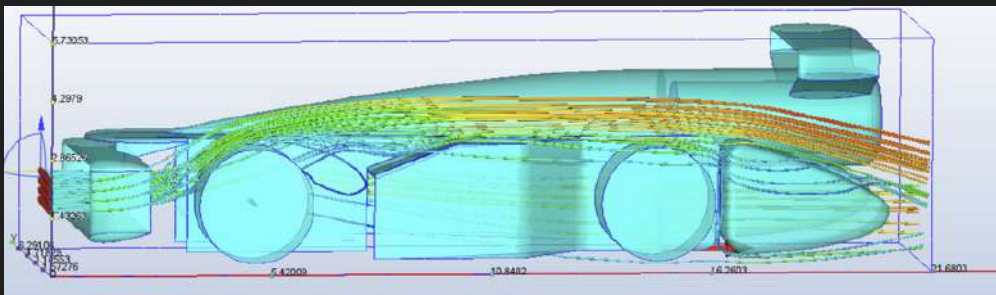
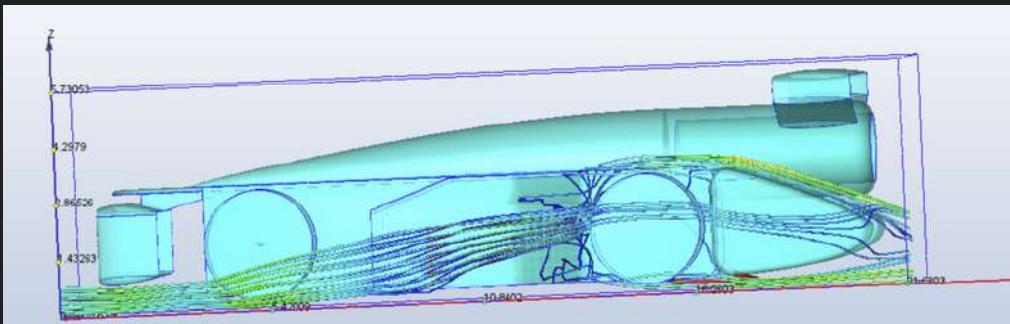
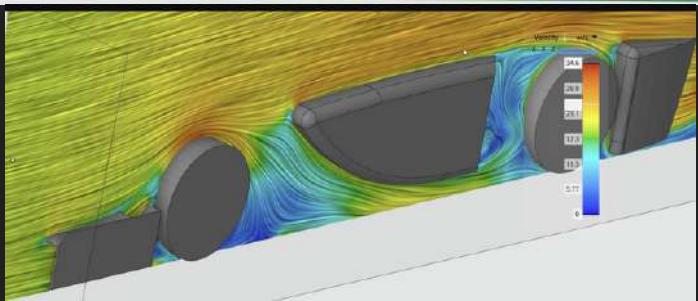
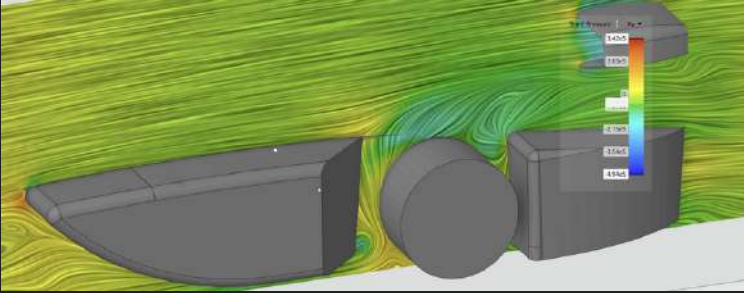
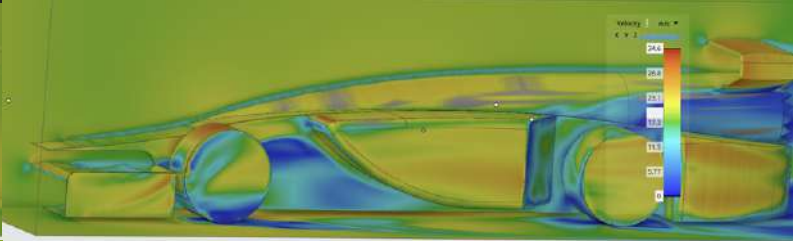
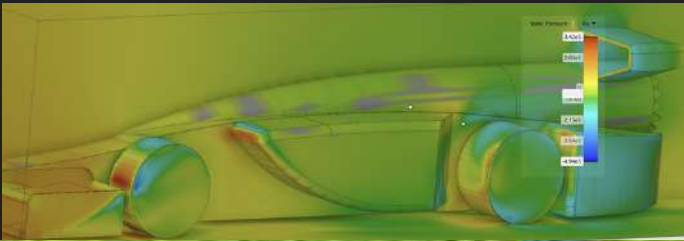
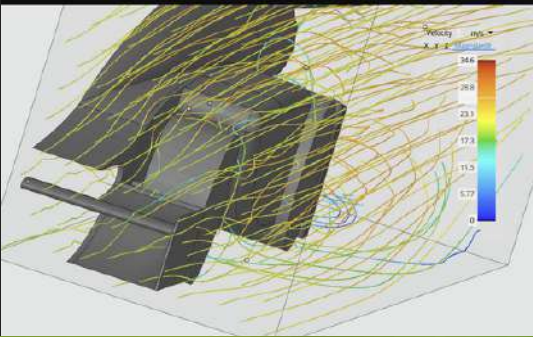
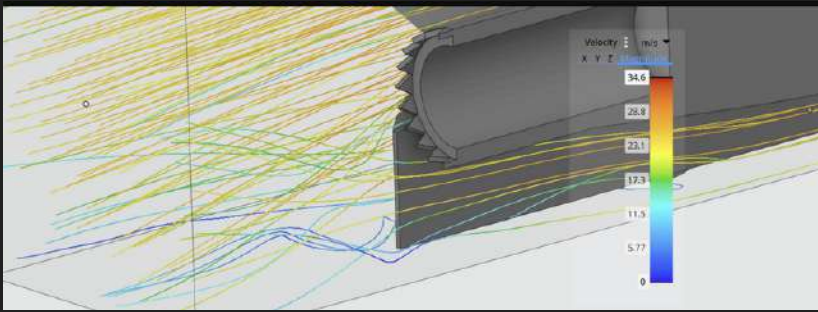


This was our produced car that we designed after continuous refinement after 38 CFD simulations.



Many of our CFD simulations failed due to mesh/geometry issues and complexities. Some provided half-results while others provided very inaccurate results.

Throughout our testing, we tried to streamline and refine areas with high drag coefficient and identify areas with turbulent airflow.



THE QUADCOPTER



Property of
Engineering Club

By Yash, Advay, and Milan

We wanted to build a quadcopter because we wanted to build something interactive, and flying things are cool.

We Can Divide the Project into 3 steps:

Step 1: Frame Assembly



The frame consists of three main parts:

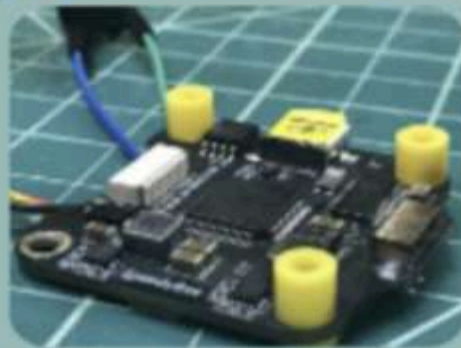
- A carbon fiber frame
- The SpeedyBee ESC and FC
- 2420 KV Brushless Motors and Propellers



Step 2: Connecting RC to ESC

Consists of:

- Radiomaster TX16 Channel Remote Controller - Transmitter.
- TBS MicroTx Receiver - soldered to ESC
- Uses Controller Area Network for transmitting signals



Flight Controller (FC)

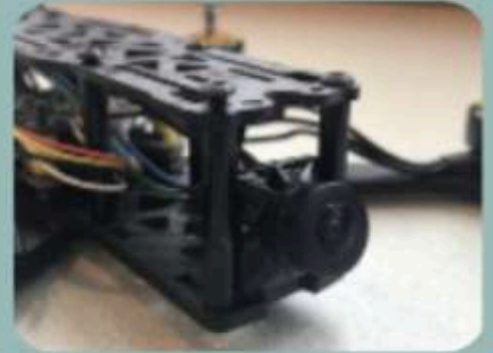


RC Transmitter

Step 3: Connecting DJI FPV to ESC

Important Specs:

- Used the DJI FPV system:
- Used the CADDX Pro Nebula Vista Camera Kit - compatible with SpeedyBee V2 ESC System and DJI Goggles
- 720p/120fps, 32 ms/4km
- FOV: 150° [D], 122° [H], 93°[V]
- Used the Low Weight of 27 grams



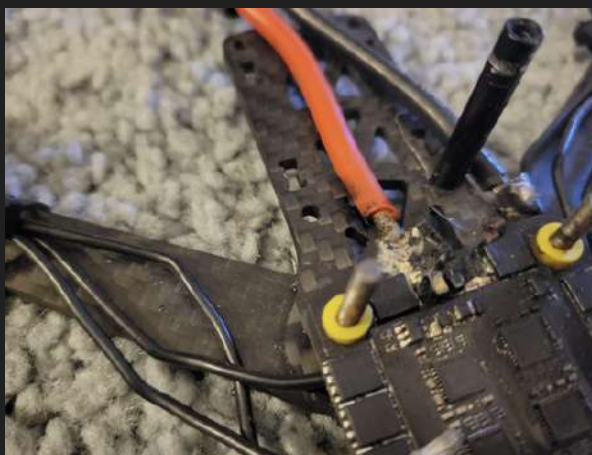
Problems

NINTENDO
GAMECUBE

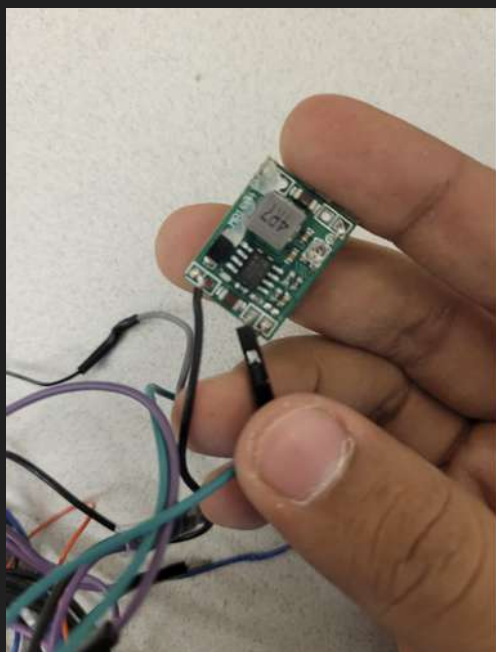
- We were completely new to this, so even with the slight details that an ideal drone builder would know, we had to do hours of research such as which batteries to use or where the video transmitter is plugged in.
- The configuration and flashing of the ESC, FC system took a lot of time due to a few manufacturing defects.
- Due to bad soldering, our first try resulted in a major battery short-circuiting causing a small fire.
- The DJI FPV system was not readily compatible with the ESC. The ESC could not recognize the FPV system.



Camera Transmitter



Shorted Battery burning the ESC. (BAD Soldering)



Soldering Gotten Better Through MUREX PID TUNING

- Setup
- Ports
- Configuration
- Power & Battery
- Failsafe
- Presets
- PID Tuning**
- Receiver
- Modes
- Adjustments
- Servos
- Motors
- OSD
- Video Transmitter
- Sensors
- Tethered Logging
- Blackbox
- CU

PID Profile Settings
Rateprofile Settings
Filter Settings

	Proportional	Integral	D Max	Derivative	Feedforward
Basic/Acro					
ROLL	56	75	50	37	202
PITCH	58	78	51	38	210
YAW	56	75	0	0	202

Mode: **RPY**

Low
Default
High

Damping: <i>D Gains</i>	1			
Tracking: <i>P & I Gains</i>	1			
Stick Response: <i>FF Gains</i>	1.35			
Dynamic Damping: <i>D Max</i>	1			
Drift - Wobble: <i>I Gains</i>	0.75			
Pitch Damping: <i>Pitch/Roll D</i>	0.9			
Pitch Tracking: <i>Pitch/Roll P, I & FF</i>	1			
Master Multiplier:	1.25			

PID Controller Settings

Feed-forward	7	Jitter Reduction
	65	Smoothness
	2 Point	Averaging
	15	Boost
	90	Max Rate Limit
	0	Transition
☒	I Term Relax	
	RP	Axes
	Setpoint	Type
	15	Cutoff
☒	Anti Gravity	
	☒	Permanently enable
	Smooth	Mode
	6	Gain
☐	I Term Rotation	
Dynamic Damping	37	Gain
	20	Advance

Throttle and Motor Settings

5	Throttle Boost
100	Motor Output Limit
30	Dynamic Idle Value [* 100 RPM]
☐	What Say Compensation